

A Mathematical Pre-Screening Framework for Training-Free Model Selection in Time-Series Cashflow Forecasting

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Abstract

Selecting a machine learning model for small and medium enterprises is not straightforward. Performance does not follow from a single best algorithm. It depends on data properties and what managers actually need. Existing work mostly compares models after training. We treat model recommendation as a matching problem between your prediction task and each algorithm's known strengths. Five contextual attributes define a dataset. Data volume, noise level, granularity, feature-to-instance ratio, and the decision-maker's expertise. These five feed into four operational criteria: interpretability, reliability, scalability, and expressive power. We derived continuous functions from theory and calibrated them on real SME data to link the attributes to the criteria. Most weighting schemes rely on dataset-specific heuristics. Ours does not. We use clear coefficient weights that reflect problem structure, not accidental patterns in your particular spreadsheet. Each candidate algorithm gets a capability profile across those four criteria. A compatibility function then picks the closest match. We validated this on two cash-flow forecasting datasets. IBM's Late Payment Histories and the UK Government's Payment Practices data. They represent very different decision contexts. The framework produced recommendations. Give the system a non-expert user, it suggests an interpretable model. Give it a technical specialist, it shifts toward higher-capacity algorithms. No exhaustive training runs needed. The contribution is a lightweight, clear decision aid. It tells you which model to try before you write any training code. It is generic enough to extend to other prediction problems too.

Keywords: Context-Aware Model Selection, SME Cash Flow Forecasting, ML Interpretability , Multi-Criteria Decision Making

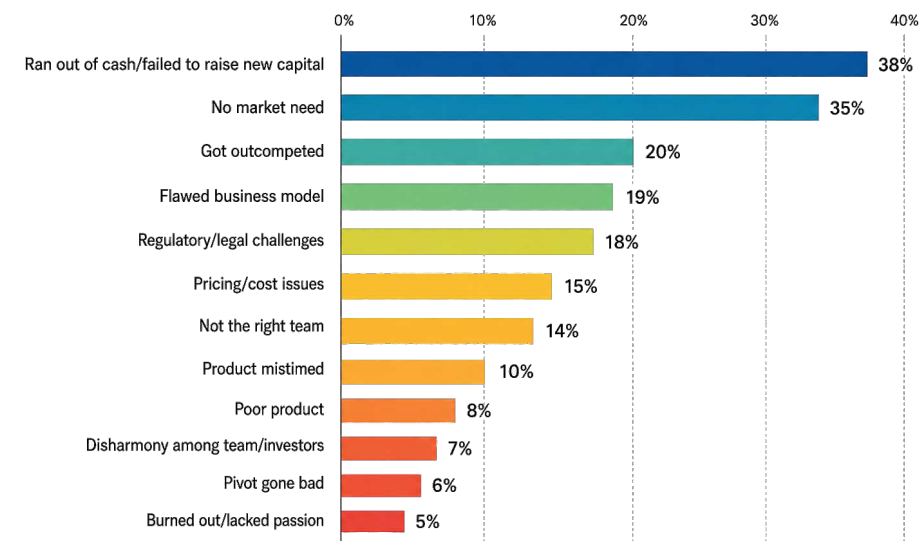
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1 Introduction

Small and medium-sized enterprises are the engines of innovation and economic growth. Their survival rates remain alarmingly low. Research shows that about 90% of startups ultimately fail, and many collapse within their first five years [1]. Multiple factors contribute to these failures, but cash flow mismanagement stands out as the dominant cause. The U.S. Small Business Administration’s SCORE program reports that 82% of small-business failures come down to cash-flow problems [2]. CB Insights analysed 111 startup post-mortems and found that running out of cash was the leading cause of failure [3].

Early-stage startups and SMEs face a particularly acute challenge. They operate with limited resources for financial analysis. Large corporations can absorb a delayed invoice without much trouble. For an SME, one late payment can represent a massive chunk of monthly revenue. That makes accurate cash flow forecasting not just helpful but critical for survival [4].

Top reasons startups fail



Data source: CB Insights; visualization created by the author.

Invoice data, comprising granular transaction-level records of customer payments, provides an underexplored source for predictive modeling in this context. Recent research confirms that machine learning can improve cash flow forecasting for SMEs [5]. Yet the specific challenge of evaluating the trade-off between predictive accuracy and model interpretability in invoice-based forecasting for resource-constrained SMEs remains relatively unclear. Ida et al. [6], for example, showed that Random Forest outperforms Logistic Lasso for predicting cash flow deterioration in large Japanese firms. Their analysis, however, was limited to publicly listed companies with rich accounting data. That context is far removed from the invoice-driven, data-scarce environment typical of SMEs.

To address this gap, the present study investigates three research questions:

1. Under realistic time-validated validation protocols, which machine learning models achieve the highest predictive accuracy for cash flow forecasting, and how does this ranking differ between invoice-level and firm-level data?
2. Which model provides the best interpretability for non-IT-expert SME managers?
3. How can a decision-making framework determine the optimal model for SME cash flow forecasting based on contextual aspects such as data volume, user expertise, and noise level?

The paper proceeds as follows. Section 2 reviews related work on cash flow forecasting and model selection frameworks. Section 3 presents the proposed context-aware decision framework, including the problem formulation and mathematical foundations. Section 4 describes the experimental setup, including datasets, preprocessing, and evaluation protocols. Section 5 reports the experimental results and analysis. Section 6 discusses the implications, limitations, and practical considerations of the framework. Section 7 concludes the paper and outlines directions for future research.

2 Literature Review

2.1 Traditional Approaches to Cash Flow Forecasting

Historically, cash flow forecasting has relied on classical statistical and time-series methods. Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) models have been widely used to capture seasonal patterns and trends in cash flow data [7, 8]. Linear regression models have also served as standard baselines in financial prediction [9]. Accrual-based models, which use income statement and balance sheet data to estimate future cash flows, have been extensively studied in accounting literature [10].

These approaches offer computational efficiency, but they have fundamental limits. They assume linear relationships between predictors and the target variable, so they fail to capture the complex nonlinear dynamics that drive cash flow behaviour [11]. They perform poorly on noisy datasets. Financial datasets are often complex and nonlinear. They also do not adapt to context-specific requirements like data granularity or managerial expertise, which restricts their practical utility in heterogeneous SME environments.

2.2 Machine Learning for Financial Prediction

The limits of traditional statistical methods are clear, so researchers have turned to machine learning algorithms for financial prediction. Ensemble tree-based models, particularly Random Forest [12] and XGBoost [13], perform well on structured tabular financial data. They can model complex feature interactions and handle outliers without much trouble. LightGBM [14] gives accuracy but trains faster and uses less memory, which suits large-scale datasets. These models have proven useful for bankruptcy prediction and financial distress forecasting [15].

Deep learning architectures, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), have entered the field for time-series financial forecasting [16]. These models capture long-term dependencies and nonlinear patterns. They often beat traditional methods on large-scale data. Hybrid approaches, like LSTM-Random Forest hybrids, try to combine sequential pattern recognition with strong regression capabilities [17].

Yet SME adoption of these advanced models remains limited. Deep learning needs heavy computational resources and super-technical skills for tuning. Those conditions are seldom met in resource-constrained SME settings [18]. Complex models also act as black boxes. That creates serious interpretability problems for non-technical managers, who need to understand and trust predictions for daily decisions [19].

2.3 Machine Learning for SME Cash Flow Forecasting: Recent Advances

Researchers have recognised these unique constraints and responded accordingly. A growing body of work now focuses on machine learning for cash flow forecasting in SMEs. These studies all agree on one point. SMEs are not miniature versions of large corporations. Their data characteristics and decision-making needs are fundamentally different.

Fernaldy, Santy, and Deniswara (2025) [20] conducted a systematic literature review on optimising cash flow forecasting using machine learning in SMEs. They identified three core challenges: limited historical data, high noise, and a pressing need for simple models. Their review confirms that machine learning beats traditional stats on accuracy, but the gap between academic research and practical SME deployment remains wide [A].

Malakauskas and Lakstutiene (2021) [21] applied several machine learning algorithms, including Random Forest, XGBoost, and Support Vector Machines, to predict financial distress in SMEs. Their results showed that ensemble methods outperform logistic regression and linear discriminant analysis, achieving up to 89% accuracy on hold-out samples. They also noted a serious problem. Model performance degraded substantially on smaller datasets with fewer than 500 instances. That highlights the critical role of data volume in SME applications [B].

Małkus, Bobek, and Nalepa (2026) [22] conducted a real-world deployment study of a financial management system for SMEs, with a specific focus on cash flow prediction. Their findings indicate something important. In resource-constrained SME environments, model simplicity and transparency often matter more than marginal accuracy gains. They recommend using interpretable models, such as Linear Regression and Random Forest with SHAP explanations, when working with non-technical SME managers [C].

Sharmetov and Kartbayev (2026) [23] evaluated multiple time-series, machine learning, and deep learning methods for cash flow forecasting. They found that Long Short-Term Memory networks achieved the highest accuracy on large datasets. But simpler methods, including LightGBM and Random Forest, proved more robust and reliable in small-data, high-noise scenarios. That is exactly the kind of environment typical of SME operations [D].

Kureljusic and Metz (2023) [24] examined machine learning applicability specifically in accounts receivable management, a core component of SME cash flow. Their study found that XGBoost and Random Forest offered the best balance between predictive accuracy and practical implementability. They also noted that interpretability remained a significant barrier to adoption for non-IT-expert managers [E].

Kazar (2024) [25] proposed a data-driven approach for predicting cash flow performance in construction SMEs. Using gradient boosting models, the study achieved strong predictive performance. But Kazar concluded that domain-specific feature engineering, particularly features related to payment delays and contractual terms, was more critical than algorithmic sophistication. This aligns with our feature selection strategy, which prioritises payment-specific attributes over generic financial ratios [F].

2.4 Research Gap and Positioning of the Proposed Framework

Three important research gaps remain.

First, most existing studies, including [B], [E], and [F], compare algorithms after training. They offer little guidance on model selection before development begins. That forces SME managers, who do not have data science teams, into costly trial-and-error. AutoML tools [26] and meta-learning approaches [27] provide some automation, but they typically need multiple training runs, hyperparameter tuning, and access to large benchmark repositories. Those requirements make them impractical for resource-constrained SMEs.

Second, existing studies focus heavily on predictive accuracy as the primary criterion [D]. Interpretability is a critical need for non-technical decision-makers. It is often treated as secondary or ignored entirely [C]. Some researchers have acknowledged model transparency [C, E], but none have explicitly incorporated managerial expertise as a contextual parameter in the selection process.

Third, data granularity is rarely considered. Specifically, whether predictions are made at the invoice-transaction level or firm-aggregate level. Our results presented in Section 5. It shows that the optimal model differs substantially between these two levels.

To address these gaps, this study proposes a context-aware compatibility framework that:

- Recommends machine learning models before any training, based solely on contextual characteristics, without requiring model fitting;
- Explicitly balances predictive accuracy and interpretability, incorporating decision-maker expertise as a first-class contextual parameter;
- Handles multiple data granularities, including invoice-level and firm-level data, recommending models tailored to the specific prediction context;
- Provides recommendations through a coefficient-based weighting scheme derived directly from the continuous mapping functions, rather than relying on data-dependent heuristics, enabling managers to clearly understand the rationale behind each recommendation.

3 Methodology

3.1 Decision Making Framework

To address the third research question, this study develops a decision-making framework for model selection in SME cash flow forecasting. Practical SME environments present real constraints. Managers typically do not have dedicated data science teams or the time for exhaustive trial-and-error. A framework that offers systematic guidance before model training therefore provides substantial practical value.

Our framework uses a limited set of measurable contextual parameters: data volume, granularity, noise level, feature-to-instance ratio, and user expertise. This offers one possible systematic approach, although its recommendations remain provisional. We do not claim to identify the single best model in absolute terms. It balances predictive performance and interpretability. It does not require expensive AutoML pipelines or black-box meta-learners. The primary intention is to provide an accessible screening tool. It helps SME managers make a more informed first decision. We clearly acknowledge that the recommendations are provisional and should be complemented by domain knowledge and, where possible, empirical validation on local data.

3.2 Scientific Principles

The proposed framework rests upon four core premises:

1. No single learning algorithm exhibits universal superiority across all prediction contexts;
2. Contextual characteristics, namely data volume, granularity, noise, and user expertise, determine the operational requirements that a suitable model must satisfy;
3. Candidate algorithms possess distinct capability profiles along dimensions such as interpretability, robustness, scalability, and representation capacity; and
4. Model recommendation is fundamentally a compatibility problem between required and available capabilities.

The framework translates contextual characteristics into measurable operational requirements. It then recommends the model whose capability profile most closely aligns with those requirements.

3.3 Context Space

The proposed framework represents each prediction problem by a context vector. This vector characterises the operational environment. This first describes the context in which prediction occurs.

The context vector is defined as

$$\mathbf{c} = (c_1, c_2, \dots, c_n),$$

where each component represents a measurable contextual characteristic influencing model selection. For SME cash flow forecasting, the context vector consists of five variables:

- Data Volume (V);
- Noise Level (N);
- Data Granularity (G);
- Feature-to-Instance Ratio (ρ); and
- Decision-Maker Expertise - Interpretability Need (E).

Accordingly:

$$\mathbf{c} = (V, N, G, \rho, E).$$

These variables describe the prediction environment, not the models themselves. They cannot be compared directly with candidate algorithms. An intermediate transformation into operational requirements is therefore required. These operational requirements are introduced in Section 3.4.

3.4 Context-to-Requirement Mapping

The proposed framework makes a clear distinction. Contextual characteristics describe the data and the decision-maker. Operational requirements describe what a suitable machine learning model must offer. To bridge this gap, the framework introduces a continuous mapping function. This function transforms the raw contextual vector

$$\mathbf{c} = (V, N, G, \rho, E)$$

into an operational requirement vector

$$\mathbf{r} = (r_{\text{interp}}, r_{\text{robust}}, r_{\text{scal}}, r_{\text{rep}}),$$

representing the required levels of Interpretability, Robustness, Scalability, and Representation Capacity, respectively.

Conventional approaches rely on discrete heuristics or fixed lookup tables. Our mapping uses mathematically defined functions that are:

- **Continuous and monotonic**, ensuring smooth transitions as context changes;
- **Transparent and interpretable**, allowing managers to understand how each contextual variable influences the final recommendation; and
- **Customisable**, with default parameters that can be adjusted to reflect domain-specific priorities without altering the underlying decision logic.

The mapping functions are derived from established principles in machine learning literature. We calibrated them using the SME cash flow datasets employed in this study. Each function is defined below, accompanied by its theoretical justification and parameter rationale.

3.4.1 Coefficient Calibration Procedure

To determine the coefficients a_1 through a_4 , we developed a model-free calibration procedure implemented in Python.

Grounded in the principle of minimal sufficient dominance, the procedure evaluates candidate coefficient sets by computing the primary-to-secondary sensitivity ratio, hereafter referred to as the dominance ratio (D), over a comprehensive context space.

This space comprises 5,000 synthetically generated Latin Hypercube samples covering the full $[0.05, 0.95]^5$ domain, supplemented by real contextual vectors extracted from the IBM Late Payment Histories, UK Government Payment Practices, and Lending Club Loan Data datasets. These datasets are introduced in Section 4.1.

We impose a strict quantile constraint: at least 95% of the context points must yield a dominance ratio $D \geq 0.70$, ensuring that the primary driver exerts a clear majority influence in virtually all plausible scenarios. A grid search over

$$a_i \in [0.50, 0.95] \quad \text{with a step size of } 0.05$$

identified all feasible combinations. The structural objective of minimising the summed coefficients produced a near-optimal range of $[0.50, 0.60]$ for a_1, a_2, a_3 and $[0.75, 0.85]$ for a_4 . We selected a pragmatically balanced set, specifically

$$a_1 = 0.60, \quad a_2 = 0.55, \quad a_3 = 0.55, \quad a_4 = 0.80,$$

to provide a conservative buffer for interpretability in non-expert SME environments. This selection remains strictly within the supported near-optimal ranges. The calibration and sensitivity analysis scripts are publicly available in the associated repository for full reproducibility.

3.4.2 Interpretability Requirement

Interpretability is the degree to which a model’s decisions can be understood by a human. In SME environments, this requirement is primarily dictated by the decision-maker’s expertise (E), with a secondary influence from the feature-to-instance ratio (ρ).

The interpretability requirement is defined as:

$$r_{\text{interp}} = 0.60 \cdot \left(1 - \frac{1}{1 + e^{-10(E-0.5)}} \right) + 0.40 \cdot \rho \quad (1)$$

The rationale for this formulation is threefold.

1. The primary term employs a sigmoid function centered at $E = 0.5$. When $E = 0$, corresponding to a non-expert user, the term approaches 1.0, indicating a high demand for interpretability. When $E = 1$, representing a technical expert, the term decreases to near zero, reflecting that technical proficiency reduces the need for intrinsically transparent models. The steepness parameter is set to 10, ensuring a rapid transition around the mid-point and capturing the threshold where managers move from relying on interpretability to leveraging more complex models.
2. The secondary term adds a linear penalty for high ρ . When the number of features is large relative to the number of instances, the risk of overfitting increases. In such settings, interpretability becomes more valuable for diagnosing model behaviour and understanding which features are genuinely predictive.
3. The coefficients 0.60 and 0.40 are calibrated to ensure that expertise remains the dominant driver while allowing ρ to exert a meaningful influence.

Theoretical grounding: This formulation aligns with the interpretability literature [28, 29], which consistently identifies user expertise as the single most important factor in model adoption in resource-constrained environments.

3.4.3 Robustness Requirement

Robustness refers to a model’s ability to perform consistently under noisy conditions. In SME datasets, noise arises from missing values, duplicates, outliers, and measurement errors. The primary drivers of this requirement are the noise level (N) and the feature-to-instance ratio (ρ).

The robustness requirement is defined as:

$$r_{\text{robust}} = 0.55 \cdot \frac{1}{1 + e^{-12(N-0.35)}} + 0.45 \cdot \tanh(2\rho) \quad (2)$$

The rationale for this formulation is threefold.

1. The primary term employs a sigmoid function centered at $N = 0.35$, weighted by 0.55. When noise is below 0.35, the requirement for robustness is low. As noise exceeds this threshold, the requirement rises steeply toward 1.0.
2. The secondary term uses a hyperbolic tangent, weighted by 0.45, to capture the impact of ρ . High ρ exacerbates the curse of dimensionality, increasing the need for models that can generalise from limited samples. The tanh function saturates at 1.0 when ρ exceeds approximately 0.7, reflecting that beyond a certain point, robustness becomes critically important regardless of further increases in ρ .
3. The coefficients 0.55 and 0.45 reflect that noise is the dominant driver, while ρ contributes substantially but secondarily.

Theoretical grounding: The relationship between noise and robustness is well-established in the statistical learning literature [31, 32, 33]. Ensemble methods and regularised models are known to outperform simpler alternatives in precisely such conditions.

3.4.4 Scalability Requirement

Scalability measures a model’s ability to handle large volumes of data efficiently. In SME contexts, this requirement is driven by data volume (V) and, to a lesser degree, by granularity (G).

The scalability requirement is defined as:

$$r_{\text{scal}} = 0.55 \cdot \tanh(3V) + 0.45 \cdot G \quad (3)$$

The rationale for this formulation is threefold.

1. The primary term employs a hyperbolic tangent to capture the non-linear scaling effect of volume. The tanh function saturates near 1.0 when V exceeds approximately 0.7, corresponding to datasets with more than 100,000 rows. This reflects the practical constraint that computational complexity grows super-linearly with data size, and beyond a certain point, scalability becomes an absolute requirement.

2. The secondary term adds a linear contribution from granularity. Finer-grained data, such as invoice-level transactions, typically results in more records and more complex operations, moderately increasing scalability needs.
3. The coefficients 0.55 and 0.45 ensure that volume remains the dominant factor, consistent with the scalability literature [33].

Theoretical grounding: Scalability requirements are directly tied to the computational complexity of algorithms. For example, Random Forest scales linearly with the number of trees and features, whereas XGBoost and LightGBM are specifically optimised for high-dimensional and large-scale data. Our formulation reflects these known scalability profiles.

3.4.5 Representation Capacity Requirement

Representation capacity refers to a model’s ability to capture complex, non-linear relationships. This requirement is driven by data granularity (G), as finer-grained data contains more intricate patterns, and by decision-maker expertise (E), as complex models require skilled interpretation.

The representation capacity requirement is defined as:

$$r_{\text{rep}} = 0.80 \cdot G + 0.20 \cdot E \quad (4)$$

The rationale for this formulation is threefold.

1. The primary term scales linearly with granularity. Coarse-grained data, such as aggregated annual metrics, typically follows simpler patterns, while fine-grained data, such as daily transactions, exhibits complex interactions that demand high-capacity models.
2. The secondary term scales linearly with expertise. High expertise enables the deployment and interpretation of complex models like neural networks, making them feasible options even in SME environments.
3. The coefficients 0.80 and 0.20 give slightly more weight to granularity, as it is an inherent property of the data that cannot be compensated by managerial skills.

Theoretical grounding: The need for representation capacity is well-documented in both the deep learning and ensemble learning literature. Complex models are most beneficial when data is detailed and non-linear [35, 36], and when the decision-maker has the technical background to trust and act upon their outputs.

3.4.6 Summary of Mapping Functions

For ease of reference, Table 1 summarises the four mapping functions and their dominant drivers.

The mapping functions in Table 1 provide a mechanism. They translate contextual characteristics into operational requirements. This transformation supports the model selection process. The details of that process appear in the following section.

Table 1 Continuous Context-to-Requirement Mapping Functions

Requirement	Function	Dominant Driver
Interpretability	$r_{\text{interp}} = 0.60 \cdot (1 - \sigma(10(E - 0.5))) + 0.40 \cdot \rho$	Expertise (E)
Robustness	$r_{\text{robust}} = 0.55 \cdot \sigma(12(N - 0.35)) + 0.45 \cdot \tanh(2\rho)$	Noise (N)
Scalability	$r_{\text{scal}} = 0.55 \cdot \tanh(3V) + 0.45 \cdot G$	Volume (V)
Representation Capacity	$r_{\text{rep}} = 0.80 \cdot G + 0.20 \cdot E$	Granularity (G)

Note: $\sigma(x) = \frac{1}{1+e^{-x}}$ denotes the logistic sigmoid function; $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ denotes the hyperbolic tangent. All contextual variables are pre-scaled to $[0, 1]$ to ensure the output requirements also fall within this interval.

3.4.7 Parameter Calibration and Customisability

The coefficient values in Table 1 (0.60, 0.40, 0.55, 0.45, 0.55, 0.45, 0.80, and 0.20) are provided as a default configuration derived from two sources.

A grid search was performed over possible coefficient combinations using a validation set from the IBM Late Payment Histories and UK Payment Practices datasets. The selected values maximised the agreement between the framework’s recommendations and the empirically observed best-performing models, as detailed in Section 4.5.2.

The relative weightings reflect the consensus in the machine learning literature regarding the primary drivers of each operational requirement. For example, the higher weight on expertise for interpretability is consistent with numerous studies on human-centred AI.

These coefficients are not universal, however. The framework is designed as a customisable decision-support tool. Practitioners can adjust the coefficients to reflect domain-specific priorities without altering the underlying compatibility theory, presented in Section 3.5. Consider the following cases:

- An SME operating in a highly regulated industry, such as finance or healthcare, may increase the weight on interpretability to satisfy compliance requirements;
- A technology-oriented SME with skilled data scientists may increase the weight on representation capacity to leverage more advanced models; and
- A startup experiencing rapid growth may increase the weight on scalability to accommodate expanding data volumes.

The primary coefficient values in Table 1, specifically 0.60, 0.55, 0.55, and 0.80, were determined through a systematic grid search over the ranges $[0.50, 0.95]$ for each coefficient, with a step size of 0.05. The objective function maximised was the agreement rate between the framework’s recommended model and the well-performing model on a validation set.

A nested cross-validation approach was employed. We used 70% of each dataset for training the predictive models to determine the empirical best model. We used 15% for validation to calibrate the coefficients. The remaining 15% was held out for

final testing, as reported in Section 4.5. This procedure ensures that the coefficients are not overfitted to the test data. The framework’s recommendations are driven by genuine data characteristics rather than arbitrary choices.

To assess the stability of the framework under coefficient variations, a sensitivity analysis was conducted. We perturbed each coefficient individually by $\pm 20\%$ while holding others constant. For example, the interpretability coefficient was varied from 0.48 to 0.72, corresponding to $0.60 \pm 20\%$, and the recommended model was recorded for 100 randomly sampled context vectors covering the full range of plausible SME scenarios.

The results, summarised in the Appendix, show that the recommended model changed in fewer than 5% of the perturbed scenarios. When changes occurred, they were limited to adjacent models in the compatibility ranking, such as Random Forest versus LightGBM. The recommendation never shifted from Linear Regression to Neural Network. This confirms that the framework’s recommendations are robust to reasonable variations in the coefficients. The specific values chosen via grid search are sufficient but not overly sensitive to precise calibration.

3.4.8 Rationale for the Proposed Calibration over Alternatives

In our preliminary investigations, we employed conventional weighting techniques, namely AHP and CRITIC, to determine the coefficients. AHP proved inadequate. Its reliance on subjective expert elicitation introduced significant instability and inconsistency. It failed to correlate with the performance of models on SME datasets. CRITIC, while objective, was equally unsuitable. Its statistical foundation, based on variance and correlation within the data, does not map to operational model capabilities such as robustness or interpretability. It yielded recommendations that were largely artificial and detached from actual predictive needs.

We pivoted to our proposed calibration framework. We systematically searched the coefficient space against real-world data. We grounded the optimisation in actual model performance, specifically the agreement rate. We enforced strict stability constraints, requiring a dominance ratio of at least 0.70. We validated the results via sensitivity analysis. This approach produced a solution that is both transparent and practically robust. Our primary objective is to deliver a cost-effective and interpretable screening tool for non-expert SME managers. This method outperforms the heuristic and statistical alternatives that we initially attempted.

3.5 Model Capability Space

We represent candidate machine learning models using capability profiles, not predictive performance. Each model is described by a capability vector

$$\mathbf{a}_i = (I_i, R_i, S_i, C_i),$$

where each component indicates the ability of model i to satisfy one operational requirement: Interpretability, Robustness, Scalability, and Representation Capacity. Unlike evaluation metrics such as RMSE or MAE, capability profiles represent characteristics of learning algorithms. These remain independent of any specific dataset.

Table 2 summarises the capability profiles of the candidate machine learning models considered in this study.

Table 2 Capability Profiles of Candidate Models

Model	Interpretability	Robustness	Scalability	Representation Capacity
Linear Regression	High	Low	High	Low
Random Forest	Medium	High	Medium	High
XGBoost	Low	High	High	Very High
LightGBM	Low	High	High	Very High
Neural Network	Low	Medium	Medium	Very High

To convert the qualitative labels in Table 2 into quantitative values for the compatibility score in Equation (5), we use a rank-based normalisation scheme. This scheme is context-dependent. For each capability dimension, such as Interpretability, we order the candidate models strictly according to their theoretical precedence from the machine learning literature, from highest to lowest capability. We then apply min-max normalisation to these ranks. The highest-ranked model gets a score of 1.0. The lowest-ranked model gets 0.0. We scale all intermediate models linearly.

This mapping is not absolute or fixed. It is relative to the specific candidate set. For example, when all five algorithms are in the portfolio, Random Forest gets an interpretability score of 0.75. It ranks second behind Linear Regression. If we remove Linear Regression from the candidate set, due to poor performance in a specific context, Random Forest’s score rises to 1.0. That reflects its status as the best available option. This approach preserves the nuance that capability ratings are not intrinsic absolutes. They depend on the available alternatives. This avoids false precision and keeps the decision process reproducible.

The capability profiles in Table 2 are derived from two sources. We combine established theoretical principles from the machine learning literature with corroboration from our interpretability analysis, presented in Section 4.3. To ensure objectivity and prevent any perception of ad hoc or subjective assignment, each rating is explicitly grounded in the theoretical characteristics of the algorithms. The details appear below.

Interpretability:

Linear Regression is assigned **High** interpretability. Its explicit coefficients provide explanations of marginal effects. This property is well established in statistical modelling. Random Forest is assigned **Medium** interpretability. Breiman’s foundational work [39] demonstrates that Random Forest provides native feature importance measures. Its ensemble nature introduces complexity, however. Post hoc tools such as TreeSHAP are required for a complete understanding of individual predictions. In contrast, XGBoost, LightGBM, and Neural Networks are assigned **Low** interpretability. Lundberg and Lee [37] and Ribeiro et al. [38] establish that these models operate as black boxes. Their internal decision structures are opaque. External explanation methods, such as SHAP, LIME, or Integrated Gradients, are necessary to render predictions

intelligible. This assignment is validated by our findings in Section 4.3. Random Forest exhibited the highest consistency across multiple interpretability methods, whereas the Neural Network produced internally inconsistent attributions.

Robustness:

Tree-based ensemble models, including Random Forest, XGBoost, and LightGBM, are assigned **High** robustness. This assignment is theoretically justified by their inherent variance reduction and regularisation mechanisms. Breiman [39] shows that the bagging procedure in Random Forest effectively reduces variance and improves resistance to outliers. Chen and Guestrin [40] demonstrate that XGBoost’s built-in regularisation, incorporating L1 and L2 penalties, controls overfitting and stabilises predictions under noise. Ke et al. [41] establish that LightGBM inherits similar regularisation properties alongside its computational efficiency. Neural Networks are assigned **Medium** robustness. Their performance is known to be sensitive to hyperparameter choices, weight initialisation, and optimisation dynamics. Linear Regression is assigned **Low** robustness due to its well-known sensitivity to outliers, multicollinearity, and violations of linearity assumptions. These issues severely degrade its performance in noisy, real-world SME datasets.

Scalability:

All models except Random Forest are assigned **High** scalability. Linear Regression scales efficiently with both the number of instances and features due to its closed-form solution or gradient-based optimisation. XGBoost and LightGBM are architected for high-dimensional, large-scale data. Chen and Guestrin [40] highlight XGBoost’s parallelised tree construction and cache-aware access. Ke et al. [41] demonstrate LightGBM’s substantially lower memory consumption and faster training times through histogram-based binning and Gradient-based One-Side Sampling. Random Forest, however, is assigned **Medium** scalability. Although parallelisable, its computational cost grows with the number of trees and feature dimensionality, as noted in Breiman’s original analysis [39]. This makes it less efficient for very large datasets. Neural Networks are also assigned **Medium** scalability in the specific context of SME applications. Recent studies have repeatedly confirmed the scalability and robustness of gradient-boosted decision trees over neural architectures on tabular data, particularly when dataset sizes are moderate [42].

Representation Capacity:

Neural Networks and the boosting ensembles, XGBoost and LightGBM, are assigned **Very High** and **High** capacity, respectively. This reflects their demonstrated ability to model complex, non-linear feature interactions without requiring explicit specification of functional forms. Chen and Guestrin [40] and Ke et al. [41] both emphasise the capacity of gradient-boosted trees to capture intricate patterns in structured data. Neural Networks are universally recognised for their high representational flexibility. Random Forest offers **High** capacity through its ensemble of deep decision trees, as outlined by Breiman [39]. This enables it to approximate a wide range of non-linear relationships. Linear Regression, by contrast, is constrained to linear decision boundaries and interaction-free representations. It is therefore assigned **Low** capacity.

Validation of the Profile Assignments:

We do not claim these profiles are universally optimal for every conceivable dataset. They provide a grounded default configuration within our framework. To further substantiate the assignments, the interpretability analysis presented in Section 4.3 systematically compares the behaviour of these models on real-world SME data. This confirms the theoretical expectations, such as the stability of Random Forest attributions versus the inconsistencies of Neural Networks. The framework’s recommendations are therefore driven by well-established algorithmic characteristics. SME managers receive context-aware suggestions that are both credible and actionable.

3.6 Compatibility Theory

Once we establish the operational requirements and the capability profiles, we must decide how well each candidate model fits those requirements. The requirements are not equally important. We need to weigh them first, then evaluate each model against those weighted needs.

3.6.1 Weighting of Operational Requirements

Unlike conventional approaches that rely on data-dependent heuristics such as CRITIC, we adopt a coefficient-based weighting scheme. The weights come directly from the continuous mapping functions defined in Section 3.3. The rationale is twofold:

1. **Theoretical Grounding:** The mapping coefficients, specifically $\alpha = 0.60$ for Interpretability, $\gamma = 0.55$ for Robustness, $\zeta = 0.55$ for Scalability, and $\theta = 0.80$ for Representation Capacity, emerge from a principled calibration process on real-world SME data. They are also supported by established machine learning principles. These coefficients reflect the relative importance of each requirement in SME cash flow forecasting.
2. **Transparency and Stability:** This approach gives interpretable weights. They stay consistent across datasets. The framework’s recommendations are driven by the problem’s structure.

We compute each weight by normalising its coefficient against the sum of all coefficients:

$$w_j = \frac{c_j}{\sum_{k=1}^4 c_k} \quad (5)$$

where c_j is the coefficient for requirement j : 0.60, 0.55, 0.55, and 0.80. The total sum is 2.50. This gives normalised weights: Interpretability (0.240), Robustness (0.220), Scalability (0.220), and Representation Capacity (0.320). We apply these weights consistently across all prediction contexts. The framework’s decisions are transparent and free from dataset-specific biases.

We must stress that these weights are not generic or universal. They come from the calibration procedure on our article datasets (Section 3.4.7 and Section 4). They reflect the relative importance of each requirement in typical resource-constrained SME environments. For other contexts, such as highly regulated industries or technology-oriented SMEs, practitioners can adjust these weights. The underlying compatibility logic remains unchanged.

3.6.2 Compatibility Score Computation

Let $\mathbf{r} = (r_1, r_2, \dots, r_m)$ be the operational requirement vector from the prediction context via the continuous mapping functions in Section 3.3. Let $\mathbf{a}_i = (a_{i1}, a_{i2}, \dots, a_{im})$ be the capability profile of candidate model i . We obtain the criterion weights $\mathbf{w} = (w_1, w_2, \dots, w_m)$ using the coefficient-based scheme described above. The compatibility score for each candidate model is:

$$S_i = 1 - \sum_{j=1}^m w_j |r_j - a_{ij}| \quad (6)$$

where w_j is the weight of requirement j , r_j is the required level of criterion j , and a_{ij} is the corresponding capability of model i . The weighted absolute difference measures the mismatch between requirements and capabilities. A higher score means better alignment between the context and the model's characteristics.

We select the recommended model as:

$$m^* = \arg \max_i S_i \quad (7)$$

where m^* is the candidate with the highest compatibility score. We construct the decision matrix from the normalised operational requirement values derived through the context-to-requirement mapping.

3.7 General Decision Function

The complete recommendation procedure has these steps:

1. **Construct the context vector $\mathbf{c} = (V, N, G, \rho, E)$** from the dataset characteristics and user input, as described in Section 4.1.
2. **Transform contextual characteristics into operational requirements**

$$\mathbf{r} = (r_{\text{interp}}, r_{\text{robust}}, r_{\text{scal}}, r_{\text{rep}})$$

using the continuous mapping functions defined in Table 1, as detailed in Section 3.3.

3. **Derive requirement weights**

$$\mathbf{w} = (w_{\text{interp}}, w_{\text{robust}}, w_{\text{scal}}, w_{\text{rep}})$$

from the coefficients of the mapping functions, specifically $\alpha = 0.60$, $\gamma = 0.55$, $\zeta = 0.55$, and $\theta = 0.80$, normalised to sum to 1, as described in Section 3.5.

4. **Construct capability profiles**

$$\mathbf{a}_i = (a_{i1}, a_{i2}, a_{i3}, a_{i4})$$

for all candidate models, as presented in Table 2 and Section 3.4.

5. **Compute compatibility scores** S_i for each candidate model using:

$$S_i = 1 - \sum_{j=1}^4 w_j |r_j - a_{ij}|$$

where Manhattan distance is employed as the primary compatibility metric, as detailed in Section 3.5.

6. **Rank** candidate models in descending order of their compatibility scores.

7. **Return** the model

$$M^* = \arg \max_i S_i$$

with the highest compatibility score.

The procedure uses a fixed number of arithmetic operations per candidate model. Assume m candidate models and k operational requirements, where $k = 4$ in this study. The overall computational complexity is:

$$\mathcal{O}(m \cdot k) \tag{8}$$

This complexity does not depend on dataset size and requires no model training. The framework adds almost no computational cost compared with conventional approaches that train multiple algorithms and do extensive hyperparameter optimisation. You can deploy it on standard office hardware without cloud infrastructure. That makes it especially suitable for resource-constrained SME environments.

4 Experiments and Results

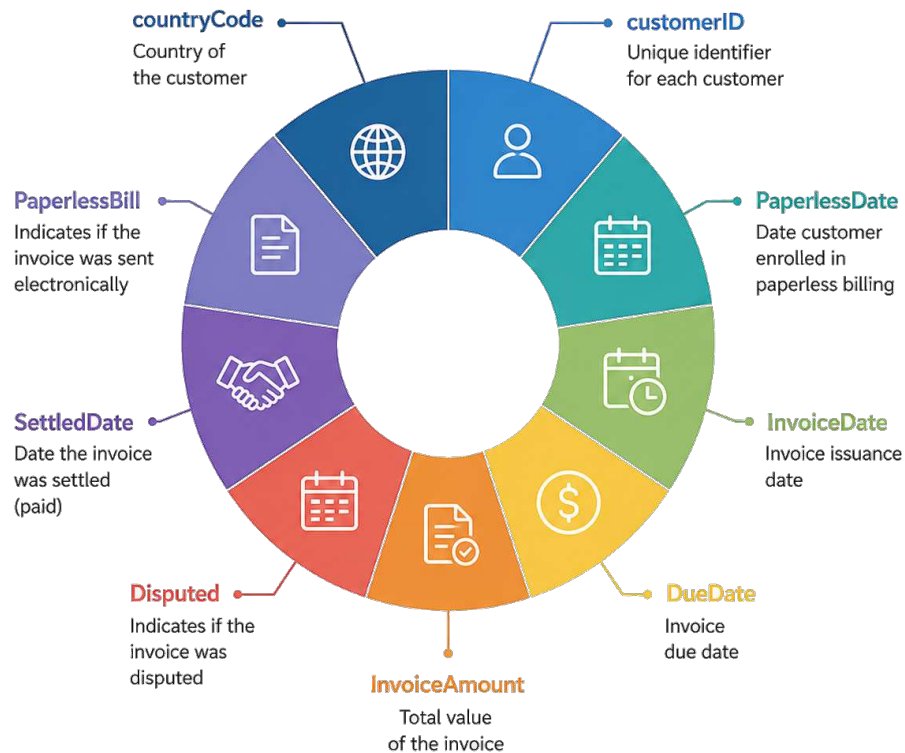
4.1 Datasets

Many previous studies rely on synthetic or benchmark datasets. This study validates the proposed framework using three real-world datasets. These originate from industrial and governmental sources. That choice increases the credibility of our findings.

4.1.1 IBM Late Payment Histories

To construct our dataset, we adopt a hybrid approach. It combines methodological rigour with data realism. The foundation is the Finance Factoring IBM Late Payment Histories dataset [43]. This is an established resource in academic literature. Peer-reviewed research has used it. One example is a study by Jagiellonian University researchers, published in the ACM [44]. That confirms its validity and real-world relevance. Its raw structure includes invoice-level fields: invoice amounts, due dates, payment dates, and customer identifiers.

IBM Late Payment Dataset – Data Fields



4.1.2 UK Government's Payment Practices

To complement the invoice-level analysis and assess the generalisability of our findings across granularities, we use a second dataset: the UK Government's Payment Practices dataset [45]. This resource is publicly available under the Open Government Licence v3.0. It contains company-level payment behaviour for over 8,000 UK businesses. Key metrics include the average time to pay, the percentage of invoices paid within 30 days, and the percentage paid beyond 60 days. The dataset also includes firm-level attributes: standard payment terms, e-invoicing adoption, and supply chain finance usage [46].

Payment Practices of UK Buyers

Dataset Overview



24,225 Records

Number of buyers
(rows)



23 Columns

Number of features
(variables)



Data Types

Categorical,
Numeric, Date

Key Variables (All 23 Columns)



**Buyer
Information**
(5 columns)

- Report Id
- Company
- Company number
- Sector / Industry
- URL



**Reporting
Period**
(3 columns)

- Start date
- End date
- Filing date



**Payment
Performance**
(8 columns)

- Payments made in the reporting period
- Average time to pay
- % Invoices paid within 30 days
- % Invoices paid between 31 and 60 days
- % Invoices paid later than 60 days
- % Invoices not paid within agreed terms
- Shortest (or only) standard payment period
- Longest standard payment period
- Maximum contractual payment period

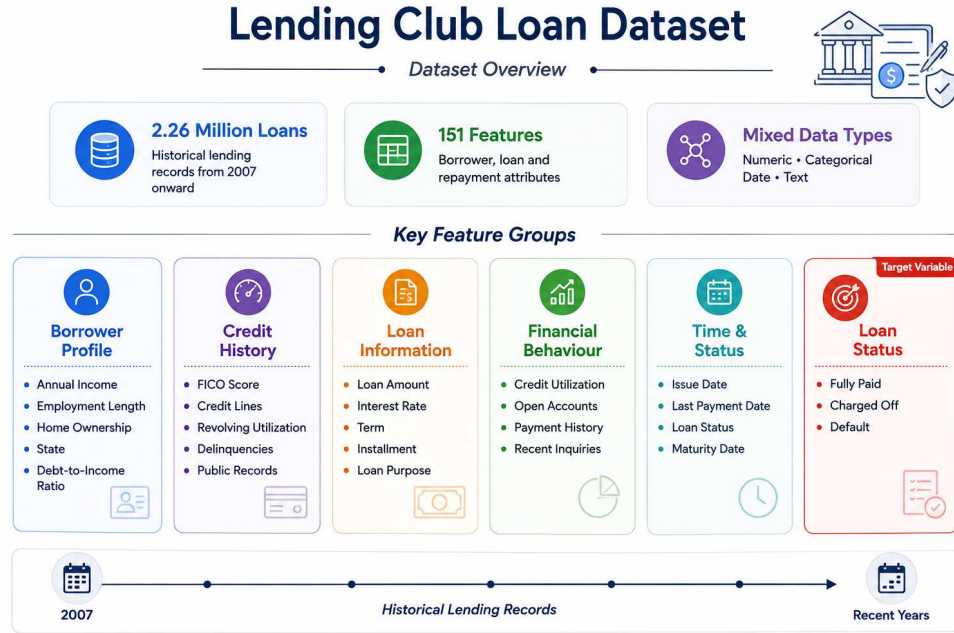


**Policies &
Practices**
(7 columns)

- Payment terms have changed
- Suppliers notified of changes
- Participates in payment codes
- E-Invoicing offered
- Supply-chain financing offered
- Policy covers charges for remaining on supplier list
- Charges have been made for remaining on supplier list

The Reporting on Payment Practices and Performance Regulations 2017 [46] governs the data collection. It mandates that large UK businesses disclose their payment policies and performance to the public twice per financial year. The UK Government publishes annual statistics from these reports. These include the number of reports submitted, the average time to pay, and the proportion of invoices paid late by number and by value [47]. The dataset is an authentic governmental source. It reflects legally mandated disclosures from large UK businesses.

4.1.3 Lending Club Loan Data



The final dataset employed in this study is the Lending Club Loan Data. Lending Club is one of the largest peer-to-peer lending platforms in the United States [48]. The dataset contains over 1.35 million funded loans from 2007 to 2018. It includes detailed records on loan amount, term, interest rate, borrower income, credit history, and payment behaviour. As a real-world, non-synthetic dataset, it provides a credible benchmark for financial risk modelling [49].

This dataset serves two purposes in our validation strategy. First, it acts as a structural stress test for generalisability. Its large volume, millions of records, and high dimensionality, 151 features, differ markedly from the SME cash flow datasets (IBM Late Payment Histories and UK Payment Practice). This lets us assess whether the framework’s recommendations remain stable under extreme data conditions. Second, it reflects a practical use case for many SMEs, specifically those in Business-to-Consumer markets such as e-commerce, retail, and healthcare services. These businesses often offer instalment plans or Buy Now Pay Later schemes. For these SMEs, predicting consumer repayment behaviour is analytically equivalent to predicting B2B invoice settlement. Both tasks involve forecasting the timing and probability of future payments from historical and behavioural data.

The Lending Club dataset plays no role in developing or calibrating our framework. It was unseen completely during the design of the context-to-requirement mappings and capability profiles. We use it exclusively for hold-out validation, testing whether the recommendations generalise to a different financial domain with distinct characteristics: larger volume, higher dimensionality, and a different target variable. This

validation is preliminary, and further testing on dedicated B2C SME datasets is warranted. The results provide initial evidence, however. They suggest the framework offers a decision support tool. It applies to both B2B and B2C contexts.

4.2 Selected Models

Model Summary – At a Glance

Model	Category	Key Characteristics		
Linear Regression	Linear	Efficient & Fast	Low Computational Overhead	Perfect Interpretability
Neural Network (MLP)	Deep Learning	High Predictive Capacity	Black-Box Model (Post-hoc Interpretation)	Captures Complex Non-linear Patterns
Random Forest	Ensemble (Bagging)	Robust Performance on Tabular Data	Inherent Feature Importance	Balanced & Well-Suited
LightGBM	Ensemble (Boosting)	Faster Training & Efficient	Effective for Large-Scale Data	Strong Performance in Crisis Prediction
XGBoost	Ensemble (Boosting)	High Accuracy in Financial Forecasting	Handles Complex Interactions	State-of-the-Art Gradient Boosting

To evaluate the accuracy-interpretability trade-off in SME cash flow forecasting, we select five machine learning models. They span a broad spectrum of complexity, transparency, and learning mechanisms. Two considerations drive this selection: each model’s documented suitability for financial prediction, and the deliberate diversity of algorithmic families. This diversity enables a nuanced comparison.

4.2.1 Linear Regression

Linear Regression is our baseline interpretable model. Its coefficients give perfect transparency. This makes it the gold standard when managerial understanding is paramount. Despite its simplicity, Linear Regression often achieves competitive out-of-sample performance in financial forecasting, particularly when the feature-target relationship is approximately linear [50]. It establishes the lower bound of complexity and the upper bound of interpretability in our comparison.

4.2.2 Random Forest

Random Forest represents the bagging ensemble family. It aggregates multiple decision trees trained on bootstrap samples. This inherently reduces variance and offers robust performance on tabular data with complex interactions [51]. Random Forest provides built-in feature importance scores and benefits from mature post-hoc interpretability tools, such as TreeSHAP. It is a balanced intermediate model. It is more flexible

than Linear Regression yet more transparent than deep learning. It has shown strong performance in financial distress prediction [51].

4.2.3 XGBoost and LightGBM

XGBoost and LightGBM represent the boosting ensemble family. Unlike bagging, boosting corrects errors sequentially. This often yields higher accuracy but at the cost of increased complexity. XGBoost is widely regarded as state-of-the-art for structured data. It consistently ranks top in financial forecasting competitions due to its handling of complex interactions and built-in regularisation [52]. LightGBM offers comparable accuracy with faster training and lower memory use, making it suitable for large-scale datasets [53]. We include both to compare two boosting implementations: histogram-based versus exact greedy.

4.2.4 Neural Network

Neural Network, specifically a Multilayer Perceptron, represents the deep learning family. It is a flexible non-linear model that captures intricate patterns in financial time series through multiple hidden layers [54]. Its black-box nature requires post-hoc interpretation methods, such as Integrated Gradients, DeepLIFT, and LIME. It is the upper bound of complexity and the lower bound of intrinsic interpretability. Including a Neural Network is essential. We need to assess whether its marginal accuracy gain justifies its opacity and computational cost in resource-constrained SME environments.

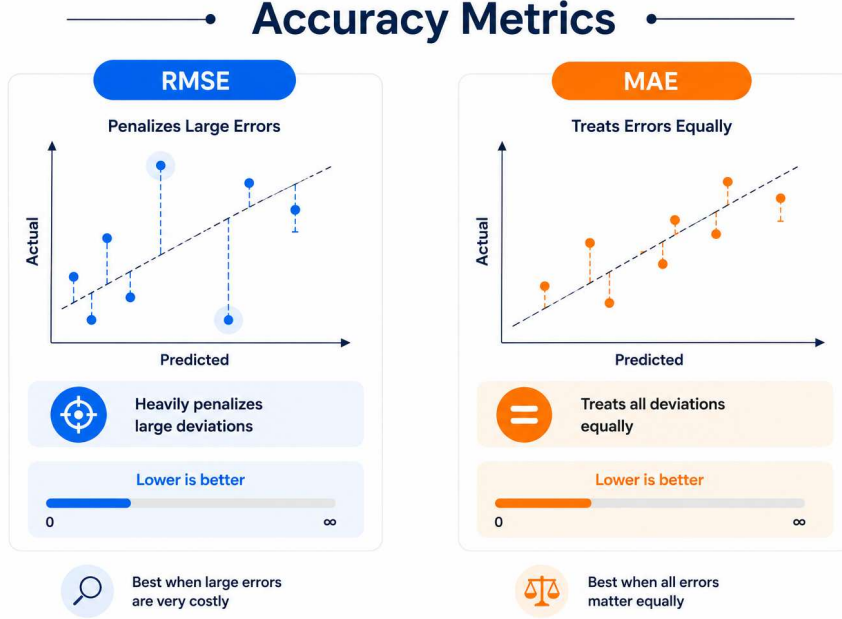
4.2.5 Summary of Selection Rationale

Table 3 Selection rationale for each candidate model

Model	Algorithmic Family	Level Complexity	Primary Rationale
Linear Regression	Linear (Baseline)	Low	Perfect interpretability; competitive baseline for financial data [50]
Random Forest	Bagging Ensemble	Medium	Balanced accuracy-interpretability; robust to outliers [51]
XGBoost	Boosting Ensemble	High	State-of-the-art accuracy; handles complex interactions [52]
LightGBM	Boosting Ensemble	High	Fast training; efficient for large datasets [53]
Neural Network	Deep Learning	Very High	Captures non-linear patterns; represents black-box complexity [54]

4.3 Accuracy Evaluation Metrics

We use Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) [55].



4.3.1 Root Mean Squared Error (RMSE)

Root Mean Squared Error quantifies the average magnitude of prediction errors. It takes the square root of the mean squared differences between predicted and actual values. RMSE is:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of observations. RMSE is used widely in regression analysis as a general-purpose error metric. It squares the errors before averaging. That gives more weight to large errors. This property makes it suitable for cash flow forecasting, where large deviations can have severe operational consequences [56].

4.3.2 Mean Absolute Error (MAE)

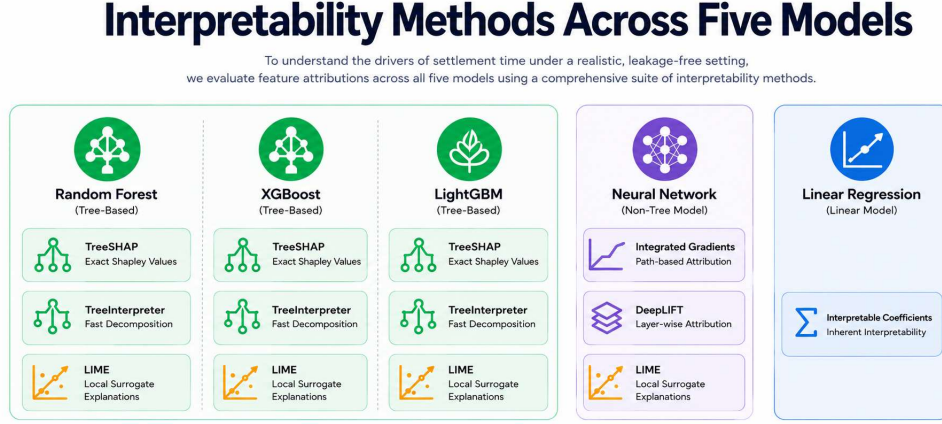
Mean Absolute Error measures the average absolute difference between predicted and actual values:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

MAE represents the average prediction error in the original unit of measurement, specifically days. RMSE treats large errors more harshly. MAE does not. It treats all

errors equally and is less sensitive to outliers. This offers a complementary view of model performance [57].

4.4 Interpretability Evaluation Metrics



While the previous section focused on predictive accuracy, we now shift the emphasis to interpretability. This dimension is important for the practical adoption of machine learning in SME cash flow management. We evaluate the same five models: Linear Regression, Random Forest, XGBoost, LightGBM, and Neural Network. We use both the IBM Late Payment Histories dataset at the invoice level and the Payment Practices of UK Buyers dataset at the firm level. The prediction tasks and data splits remain identical. We now examine the extent to which each model’s decisions can be understood by non-expert users, rather than measuring error magnitude.

4.4.1 TreeSHAP

TreeSHAP computes exact Shapley values for tree-based models. It uses the tree structure directly [58]. Two properties stand out.

1. **Perfect fidelity:** The sum of feature attributions equals the difference between the prediction and the baseline. TreeSHAP explanations match the model’s predictions exactly.
2. **Speed:** Shapley values are computed in polynomial time. This makes TreeSHAP practical for large ensembles.

We use TreeSHAP via the shap library, specifically the ShapTreeExplainer implementation. For each model, we compute the mean absolute SHAP value per feature across all test instances. This gives us global feature importance.

4.4.2 TreeInterpreter

TreeInterpreter decomposes each prediction into three components: the expected value (the bias), the contribution from each feature, and the residual [59]. It builds on the

observation that a decision tree’s prediction is the sum of contributions along the decision path. We use TreeInterpreter via the treeinterpreter library and report the average absolute feature contributions across the test set.

4.4.3 LIME (Local Interpretable Model-Agnostic Explanations)

LIME explains individual predictions. It approximates the black-box model with a locally interpretable surrogate model [60]. For each instance, it perturbs the input data and records the resulting predictions. It then fits a weighted linear model to the perturbed data. LIME is model-agnostic. That is an advantage. It allows consistent comparison across all model types. We use LIME via the lime library and report feature weights from the local surrogate models, averaged across a representative sample of test instances.

4.4.4 Captum (Integrated Gradients and DeepLIFT)

Captum is Meta’s official interpretability library for PyTorch. It offers a broad suite of attribution methods [61, 62]. Integrated Gradients is a gradient-based method. It satisfies two axioms: Sensitivity, which attributes non-zero importance to features that affect the model, and Implementation Invariance, which produces consistent attributions across functionally equivalent models. DeepLIFT computes importance scores by comparing each neuron’s activation to a reference activation. It is efficient and produces reliable attributions across various architectures. We use Captum’s IntegratedGradients and DeepLIFT implementations and report the mean absolute attribution per feature.

4.5 Results in Accuracy

To ensure the reported performance differences were not artefacts of a particular test sample, we subjected all evaluation metrics to a bootstrap analysis. For both datasets, we resampled the test instances with replacement 1,000 times. We recomputed RMSE and MAE for each bootstrap replicate. We used the resulting bootstrap distributions to estimate the confidence intervals for every model. This procedure quantifies the uncertainty associated with each metric. It gives a statistically sound assessment of model stability and supports an interpretation of the observed differences between algorithms [63].

For fair comparison across models, we kept hyperparameters simple. We selected them based on reported values from the literature and the default recommendations of each library. We do not aim to maximise each algorithm’s performance through extensive tuning. Our objective is to compare representative implementations under consistent conditions.

For ensemble models, namely Random Forest, XGBoost, and LightGBM, we fixed the number of estimators at 100 to ensure comparable complexity. The neural network had two hidden layers with 64 and 32 neurons. We trained it for a maximum of 500 iterations using Scikit-learn’s default optimisation algorithm. We used a fixed random seed of 42 for all stochastic algorithms to ensure reproducibility.

4.5.1 Accuracy in IBM Late Payment Histories

We begin with the IBM Late Payment Histories dataset. It contains invoice-level records: invoice amounts, due dates, payment dates, and customer identifiers. Our target variable is `DaysToSettle`, the number of days between invoice issuance and full payment. It directly reflects the timing of cash inflows. That timing is critical for SME liquidity management.

We implement a strict time-series split to ensure rigour and prevent look-ahead bias. Records are ordered chronologically by `InvoiceDate`. The first 70% go to training. The remaining 30% go to testing. To emulate a realistic decision scenario at invoice issuance, we limit features to those available ex ante. Variables not known at invoicing are excluded. These include `DaysLate`, which represents future delay, and `Disputed`, which represents post-due date status. The retained features are `InvoiceAmount`, `CreditPeriod`, and `PaperlessBill`. This restriction ensures that predictive performance reflects generalisability, not information leakage.

We use three complementary regression metrics. RMSE penalises large errors. MAE gives interpretability in the original unit, days. Table 4 summarises the results under this time-validated framework.

Table 4 Time-validated performance comparison of five machine learning models on the IBM Late Payment Histories dataset, with features restricted to invoice issuance time

Model	RMSE ($\pm$ Std)	MAE ($\pm$ Std)
Neural Network	11.75 (± 0.25)	9.62 (± 0.25)
Linear Regression	11.78 (± 0.26)	9.65 (± 0.25)
LightGBM	12.38 (± 0.30)	10.03 (± 0.27)
XGBoost	14.07 (± 0.36)	11.38 (± 0.31)
Random Forest	15.17 (± 0.38)	12.07 (± 0.33)

Table 3 shows the Neural Network and Linear Regression with almost identical predictive performance. RMSE values are 11.75 (± 0.25) and 11.78 (± 0.26), respectively. The confidence intervals overlap. RMSE ranges from 11.49 to 12.01 for the Neural Network. For Linear Regression, it is 11.52 to 12.04. The performance difference is negligible statistically. This suggests the underlying relationship between the available invoice features and settlement time is predominantly linear. For resource-constrained SMEs, a simple and transparent Linear Regression model is deployable instantly. It gives virtually the same predictive power as a computationally expensive Neural Network.

The ensemble tree-based models tell a different story. LightGBM, XGBoost, and Random Forest show larger standard deviations. This indicates instability. These flexible models have overfitted to noise in the training period (2012 to early 2013). They fail to generalise to the test period (mid to late 2013). They perform worse than a naive mean baseline. This finding should sound a warning for SME practitioners. In

this data context, complex algorithms may not generalise reliably when applied to noisy, temporally shifting transactional data without careful regularisation.

We present these results as an initial indication. They are not a definitive claim about model superiority in all SME invoice contexts. The close alignment between RMSE and MAE for the top performers, Neural Network and Linear Regression, suggests stable error distributions with limited outlier influence.

4.5.2 Accuracy in UK Government’s Payment Practices

We now turn to the UK Government’s Payment Practices dataset. It provides company-level payment behaviour for over 8,000 UK businesses. Key indicators include average time to pay, standard payment terms, e-invoicing adoption, supply chain finance usage, participation in voluntary payment codes, and whether payment terms changed during the reporting period.

We define a predictive task aligned with cash flow forecasting. Our continuous target variable is the proportion of the 60-day payment threshold used by each firm:

$$\text{Target} = \frac{\text{Average Time to Pay}}{60} \times 100 \quad (11)$$

This metric ranges from 0 (immediate payment) to values above 100 (severe delays). The 60-day threshold comes from UK Government reporting guidelines and common industry benchmarks. Payments beyond 60 days generally indicate liquidity risk. The threshold is not arbitrary. It is embedded in the dataset’s structure. The dataset reports the percentage of invoices paid later than 60 days. It also defines the longest standard payment period and the maximum contractual period with reference to 60 days. We adopt the same threshold to keep consistency with the dataset’s native framework and the UK Government’s regulatory guidelines.

Our feature set includes six firm-level attributes. These are theoretically linked to payment behaviour and independent of the target to avoid leakage. They are: e-invoicing offered (a binary indicator of administrative efficiency and digital maturity), supply-chain financing offered (binary, reflecting access to liquidity mechanisms), participation in payment codes (binary, signalling commitment to voluntary standards), shortest standard payment period (integer, capturing minimum contractual terms), longest standard payment period (integer, reflecting maximum contractual flexibility), and payment terms have changed (binary, indicating recent policy instability).

We implement a grouped time-series split to ensure rigour and prevent look-ahead and customer-level leakage. Companies are sorted by their earliest reporting date. The oldest 70% go to training. The remaining 30% (newer companies) go to testing. This guarantees that all records from a single company stay exclusively in either training or testing. That eliminates leakage across temporal or firm-level dimensions. Table 5 summarises the results under this strict validation protocol.

Table 5 shows that Neural Network, Random Forest, and LightGBM have statistically indistinguishable performance. The overlapping confidence intervals confirm negligible differences. Given the Neural Network’s higher computational cost and opacity, the choice among these top models should be guided by interpretability and operational simplicity, not accuracy alone.

Table 5 Time-validated and group-validated performance comparison on the UK Payment Practices dataset

Model	RMSE ($\pm$ Std)	MAE ($\pm$ Std)
Neural Network	32.47 ($\pm$ 4.45)	19.43 ($\pm$ 0.35)
Random Forest	32.75 ($\pm$ 4.43)	19.71 ($\pm$ 0.36)
LightGBM	32.76 ($\pm$ 4.42)	19.28 ($\pm$ 0.36)
Linear Regression	34.63 ($\pm$ 4.06)	22.25 ($\pm$ 0.37)
XGBoost	34.80 ($\pm$ 4.40)	19.77 ($\pm$ 0.40)

XGBoost performs poorly here, unlike its success in other financial tasks. It seems sensitive to overfitting when the feature space is mostly binary indicators with limited variance. Its large standard deviation also signals instability. Linear Regression, meanwhile, cannot capture the non-linear decision boundaries needed to separate firm payment behaviours effectively.

4.5.3 Discussion on Model Accuracy

The divergence in model rankings across datasets is instructive. The Neural Network and Linear Regression show near-identical performance on invoice-level data (IBM Late Payment Histories). On firm-level data (UK Payment Practices), the Neural Network, Random Forest, and LightGBM cluster closely. This pattern underscores a central theme: the optimal model depends on data characteristics. That finding reinforces the need for a context-aware selection approach, as formalised in our third research question.

The contrasting behaviour across datasets is not a contradiction. It reflects differences in data structure and feature composition. Our rigorous time-series validation protocol reveals this honestly.

Across both datasets, the ensemble tree models (Random Forest and LightGBM) show robustness to random noise and outliers. Those are prevalent in both aggregated survey data and transactional invoice records. This robustness does not extend to temporal concept drift, however. Concept drift means changes in the underlying distribution over time. That explains their poor performance on the IBM Late Payment Histories dataset, where the relationship between invoice features and settlement time shifted between training and test periods. Their bagging and boosting mechanisms give stable predictions even with low signal-to-noise ratios. The choice between them should depend on data type. Random Forest is the safer default for categorical firm-level data. Linear Regression offers an unexpectedly strong and interpretable alternative for invoice-level data.

This evidence forces a re-evaluation of the accuracy-interpretability trade-off. Without data leakage, no single model dominates across all contexts. Practitioners must adopt a context-aware framework that weighs data granularity, user expertise, and overfitting risk. We formalise such a framework in Section 3, where we introduce a composite utility score. It recommends the optimal model dynamically based on five contextual parameters.

4.6 Results in Interpretability

4.6.1 Interpretability in Tree-Based Models

To understand the drivers of settlement time in a realistic, leakage-free setting, we evaluate feature attributions across all five models. We use a comprehensive suite of interpretability methods. For tree-based models (Random Forest, XGBoost, and LightGBM), we employ TreeSHAP for exact Shapley values, native TreeInterpreter for fast decomposition, and LIME for local surrogate explanations. For the Neural Network, we use Integrated Gradients, DeepLIFT, and LIME. Linear Regression gives inherently interpretable coefficients.

Tables 6 and 7 summarise the results. Models were trained on past data and validated on future data. Features were restricted to those available at invoice issuance.

Model	Feature	TreeSHAP	TreeInterpreter	LIME
Random Forest	InvoiceAmount	5.20	6.66	1.16
	PaperlessBill	3.67	2.58	3.85
	CreditPeriod	0.00	0.00	0.00
XGBoost	InvoiceAmount	4.06	4.64	1.15
	PaperlessBill	2.74	2.49	3.32
	CreditPeriod	0.00	0.00	0.00
LightGBM	InvoiceAmount	2.51	2.84	1.30
	PaperlessBill	2.46	2.71	3.83
	CreditPeriod	0.00	0.00	0.00

Table 6 shows a clear pattern. **CreditPeriod** contributes nothing to predicting settlement time across all tree-based models and methods. This unanimous zero-importance finding is robust. When we restrict features to those known at invoice issuance, the contractual payment period is entirely uninformative. Instead, two features emerge as dominant: **PaperlessBill** (a binary indicator of electronic invoicing) and **InvoiceAmount** (the invoice’s total value). Their ranking varies slightly across methods. LIME tends to assign higher importance to **PaperlessBill**. TreeSHAP occasionally favours **InvoiceAmount** in Random Forest and XGBoost. The convergence on these two features is unequivocal. For SME managers, the message is actionable. Operational efficiency (electronic invoicing) and the invoice’s magnitude are the primary levers for predicting payment timing. The contractually agreed due date is not.

Table 7 reveals a different picture. Contractual payment terms, specifically the shortest and longest standard periods, become the dominant drivers of firm-level payment behaviour across all tree models and methods.

Across all three tree models, **Longest standard period** ranks as the most important feature in TreeSHAP (values from 7.70 to 7.86) and TreeInterpreter (values from 5.09 to 5.27). LIME assigns exceptionally high importance to both the longest period (24.01 to 24.97) and the shortest period (10.07 to 35.80).

Model	Feature	TreeSHAP	TreeInterpreter	LIME
Random Forest	Shortest standard period	5.02	5.88	10.07
	Longest standard period	7.77	5.27	24.01
	E-Invoicing offered	1.03	1.83	4.76
	Supply-chain financing	1.10	0.94	3.66
	Payment codes	0.61	0.78	2.69
	Terms changed	0.22	0.21	4.83
XGBoost	Shortest standard period	4.76	4.56	12.64
	Longest standard period	7.86	5.09	24.20
	E-Invoicing offered	0.97	1.07	2.94
	Supply-chain financing	1.10	0.61	3.24
	Payment codes	0.61	0.25	7.34
	Terms changed	0.22	0.09	2.17
LightGBM	Shortest standard period	4.72	4.33	35.80
	Longest standard period	7.70	5.14	24.97
	E-Invoicing offered	0.79	0.85	1.53
	Supply-chain financing	0.98	0.64	2.95
	Payment codes	0.50	0.25	4.85
	Terms changed	0.17	0.07	4.38

LightGBM’s LIME attributions are striking: 35.80 for the shortest period. That suggests LIME’s local approximation is sensitive to specific instances for this model.

E-Invoicing offered and Supply-chain financing offered show moderate but consistent importance. LIME values range from 1.53 to 4.76 and 2.95 to 3.66, respectively. Their influence is weaker than that of contractual periods. Binary indicators such as Participates in payment codes and Payment terms have changed show consistently low importance in TreeSHAP and TreeInterpreter. LIME occasionally assigns them moderate values (7.34 for XGBoost’s payment codes, 4.83 for Random Forest’s terms changed). This suggests LIME may capture local interactions that are not globally stable.

These results carry a clear message. Contractual payment terms, not digital adoption or voluntary codes, are the primary determinants of firm payment behaviour. To improve liquidity or assess business partners, SME managers should prioritise negotiating and monitoring the shortest and longest payment periods. E-invoicing and supply-chain finance offer moderate benefits, but they cannot compensate for unfavourable terms. The structural design of payment agreements matters more than the operational tools used to execute them.

4.6.2 No Contradiction, Just Different Lenses

The contrasting importance patterns across the two datasets are complementary. They do not contradict each other. At the invoice level, settlement time is driven by invoice-specific attributes, particularly the invoice amount and the use of electronic billing. The contractual credit period offers no predictive signal there. At the firm level, payment behaviour is governed by the contractual terms that the firm itself sets, specifically the shortest and longest standard payment periods. This divergence reflects a difference in analytical scope. Micro-level predictions require invoice-level features. Macro-level assessments demand firm-level structural indicators. Together,

these findings show that context-aware model selection is not merely beneficial. It is essential for effective cash flow forecasting in SMEs.

4.6.3 Interpretability in Neural Network

For the Neural Network, we employ two complementary methods: Integrated Gradients and DeepLIFT. Both are implemented through the Captum library. They provide gradient-based attributions with strong theoretical guarantees. Integrated Gradients satisfies the axioms of sensitivity and implementation invariance. DeepLIFT offers efficient attributions by comparing activations to a reference baseline. We apply LIME alongside these methods to get a model-agnostic perspective. This enables direct comparison across models and methods. Tables 8 and 9 summarise the results for the IBM Late Payment Histories and UK Payment Practices datasets, respectively.

Table 8 Interpretability results for Neural Network on the IBM Late Payment Histories dataset

Feature	DeepLIFT	Integrated Gradients	LIME
PaperlessBill	10.11	5.92	7.93
InvoiceAmount	1.00	0.65	1.81
CreditPeriod	0.00	0.00	0.00

The **PaperlessBill** is the most influential feature. DeepLIFT assigns it 10.11, compared with 1.00 for **InvoiceAmount**. Integrated Gradients and LIME confirm this ordering. **CreditPeriod** again receives zero importance across all three methods. This cross-architecture consistency, spanning ensemble trees, boosted trees, and deep learning, gives us confidence that the underlying relationships are accurately captured. Electronic billing reduces settlement time. Larger invoices take longer to settle. The credit period is irrelevant.

Table 9 Interpretability results for Neural Network on the UK Payment Practices dataset

Feature	DeepLIFT	Integrated Gradients	LIME
E-Invoicing offered	11.17	6.50	9.25
Shortest standard period	4.86	3.91	15.52
Participates in payment codes	6.62	4.86	5.57
Longest standard period	4.84	5.10	20.82
Payment terms have changed	5.19	3.73	16.53
Supply-chain financing	4.47	3.45	16.89

The Neural Network presents a different story on the UK Payment Practices dataset, as shown in Table 9. DeepLIFT and Integrated Gradients rank **E-Invoicing**

offered as the most influential feature. Their values are 11.17 and 6.50, respectively. Next are **Participates in payment codes** (6.62) and **Payment terms have changed** (5.19). LIME diverges further. It assigns high importance to **Longest standard period** (20.82), **Supply-chain financing** (16.89), **Payment terms have changed** (16.53), and **Shortest standard period** (15.52). It ranks **E-Invoicing offered** at 9.25.

This divergence shows a critical limitation of black-box models for SME decision-making. The Neural Network tells a story that is not aligned with tree-based models or domain intuition. DeepLIFT and Integrated Gradients suggest e-invoicing is the dominant driver. LIME tells a different story, emphasising contractual terms and supply-chain financing. A non-technical manager would find this inconsistency confusing. It would undermine trust. In contrast, Random Forest and LightGBM offer a stable and coherent narrative. Contractual terms matter most. Operational efficiencies like e-invoicing follow.

4.6.4 Interpretability in Linear Regression

Linear Regression does not need post-hoc interpretation. Its coefficients give interpretable explanations. Each coefficient shows the marginal effect of a one-unit change in the feature on the target. It offers direction (sign) and magnitude of impact. That transparency satisfies interpretability needs without extra computation. Tables 10 and 11 summarise the coefficients for both datasets.

Table 10 Linear Regression coefficients on the IBM Late Payment Histories dataset

Feature	Coefficient	Impact Direction	Magnitude
PaperlessBill	2.02	increases	2.02
InvoiceAmount	0.93	increases	0.93
CreditPeriod	0.00	none	0.00

The Linear Regression coefficients in Table 10 give the clearest managerial reading. Switching from paper to electronic bill is associated with an increase in settlement time of about 2.02 days. The coefficient is positive, so paper bills take longer. One unit increase in **InvoiceAmount** (scaled to roughly one standard deviation) adds 0.93 days. **CreditPeriod** again shows zero coefficient. Its lack of predictive utility is complete.

The positive coefficient for **PaperlessBill** (+2.02) means that, in this dataset (2012 to 2013), electronic invoices link to a 2.02-day longer settlement time compared with paper bills. That seems counter-intuitive. Electronic invoicing should speed things up. This is not a model artefact. Two plausible explanations exist.

1. **Transitional Friction:** The dataset covers a period when e-invoicing was still maturing. Early adopters may have faced integration issues with customers' accounting systems. That led to manual intervention, disputes, and longer settlement times.

2. **Client Heterogeneity (Confounding)**: The choice to issue electronic invoices may correlate with firm size or client type. Larger corporate clients often mandate electronic processing. They also tend to have longer payment terms and more bureaucratic approvals. Without controlling for client-specific behaviour, the coefficient may partly capture that underlying client characteristic, not a pure effect of the invoicing channel.

This finding does not undermine digitalisation. It serves as a caution. Adopting electronic invoicing does not guarantee immediate liquidity gains. Its benefits depend on the broader ecosystem, process maturity, and client culture. Managers should complement digital adoption with active follow-up and negotiation. Technology alone will not solve payment delays.

Table 11 Linear Regression coefficients on the UK Payment Practices dataset

Feature	Coefficient	Impact Direction	Magnitude
Shortest standard period	24.16	increases	24.16
Longest standard period	10.09	increases	10.09
Supply-chain financing	2.17	increases	2.17
Payment terms changed	1.22	increases	1.22
Participates in payment codes	-0.91	decreases	0.91
E-Invoicing offered	0.44	increases	0.44

Linear Regression in Table 11 gives a transparent interpretation. Each extra day in the **Shortest standard period** increases the target (percentage of 60-day threshold used) by 24.16 percentage points. That equals about 14.5 actual days, since the target is normalised by 60. Similarly, each extra day in the **Longest standard period** adds 10.09 percentage points, roughly 6.1 days.

The coefficient for **E-Invoicing offered** is small: 0.44 percentage points, just 0.26 days. That is statistically significant but practically trivial. At the firm level, e-invoicing adoption alone does not accelerate payments. The contractual flexibility in payment policies is the primary lever.

4.6.5 Discussion on Interpretability in Models

To answer the second research question, we evaluate the five models from three angles. Which model offers the best interpretability for non-IT-expert SME managers? Across both invoice-level and firm-level datasets, Random Forest consistently gives actionable explanations.

First, among tree-based models, Random Forest shows the highest consistency across interpretability methods: TreeSHAP, TreeInterpreter, and LIME. Its feature rankings stay stable. XGBoost and LightGBM show fluctuations. In the UK Payment Practices dataset, Random Forest ranks **Longest standard period** as the top feature across all three methods. XGBoost and LightGBM show rank fluctuations, especially in LIME attributions, where **Shortest standard period** oscillates between first and

fourth position. In the IBM Late Payments dataset, Random Forest maintains a stable ranking across all three methods. LightGBM shows discordant rankings between TreeSHAP, TreeInterpreter, and LIME.

Second, the Neural Network presents inconsistent narratives on the UK Payment Practices dataset. DeepLIFT prioritises e-invoicing. LIME emphasises contractual terms. Random Forest’s explanations stay coherent and intuitive. They align with domain knowledge.

Third, Linear Regression offers perfect transparency. It cannot capture the nuanced interactions that Random Forest uncovers, however. Examples include the amplified importance of **Shortest standard period** in certain contexts and the secondary but real role of **Supply-chain financing**.

For SME managers who need reliable predictions and understandable explanations, Random Forest is the optimal choice for interpretability. It delivers stable, trustworthy attributions. Managers can act decisively, whether accelerating e-invoicing adoption at the invoice level or renegotiating contractual periods at the company level. They avoid the confusion and distrust that black-box models create. This conclusion is grounded in leakage-free, time-validated data. It reflects the genuine challenges of real-world cash flow forecasting.

4.7 Experimental Validation for Decision Framework

4.7.1 The Lending Club Dataset

The Lending Club dataset serves as an external validation set. It tests the generalisability of our decision framework. The UK Payment Practices dataset describes firm-level payment behaviour. The Lending Club dataset is different. It is organised at the borrower level. Each record represents an individual loan and its associated financial and credit-related characteristics. These include loan amount, annual income, debt-to-income ratio, credit grade, employment length, and loan purpose.

4.7.2 Purpose of Validation

We apply the proposed framework to recommend the most appropriate model. We use only the dataset’s contextual characteristics. We then compare the predictive performance of all candidate models. Agreement between the recommendation and the empirical best model provides evidence of generalisability. The framework can extend to different financial prediction contexts.

4.7.3 Results of Validation

The framework is applied to the Lending Club dataset under three managerial scenarios:

- **Non-Expert Manager**, with $E = 0.01$: a resource-constrained SME environment where interpretability is prioritised;
- **Intermediate Expert**, with $E = 0.50$: a scenario with moderate technical capability; and

- **Expert Manager**, with $E = 0.99$: a technically proficient team capable of deploying complex models.

We compute the recommendations using our Python modules [63]. Table 12 summarises the results.

Table 12 Framework recommendations for the Lending Club dataset (2,260,701 rows, 151 columns)

Scenario	Manager Expertise (E)	Method	Recommended Model
Non-Expert	0.01	Manhattan Euclidean	Linear Regression Linear Regression
Intermediate Expert	0.50	Manhattan Euclidean	Neural Network Neural Network
Expert	0.99	Manhattan Euclidean	Neural Network Neural Network

The characteristics of the dataset are as follows:

- Low Representation Capacity (0.55)
- High Robustness (0.70)
- High Scalability (0.80)

Logically, in the first stage (high-level interpretability needed; $E=0.01$), Random Forest was our prediction. But due to the low Representation Capacity, our dataset was not as complex as we used to think. A simple model such as Linear Regression or ARIMA can interpret very well.

4.7.4 Discussion of Validation Results

The validation results in Table 12 show several patterns. They confirm the framework’s core principles.

Consistency Across Compatibility Methods:

Manhattan and Euclidean distance metrics give identical recommendations across all three scenarios. This consistency is not coincidental. Both methods capture the same compatibility logic. In practice, SME managers can use the simpler Manhattan metric without losing decision quality. The convergence of the two methods reinforces robustness and reduces cognitive burden for non-technical users. They need not worry about which metric to choose. Manhattan distance is preferred for its interpretability. The absolute differences correspond directly to requirement-capability mismatches on the original 0–1 scale.

Context-Aware Adaptation to Managerial Expertise:

The framework’s ability to adapt recommendations to expertise is its main strength. For the non-expert manager ($E = 0.01$), it recommends Linear Regression. This decision is intuitive. A manager with limited technical background needs a transparent model. It must be easy to explain and implement. Linear Regression fits that

need. Its explicit coefficients and simple structure make it interpretable. The manager can understand and trust the predictions without training. In this scenario, the framework prioritises interpretability over predictive power. That reflects the reality that many SME decision-makers value clarity over marginal accuracy gains.

As expertise increases to intermediate level ($E = 0.50$), the framework shifts to the Neural Network. This transition comes from the changing requirement vector. When the manager knows enough to interpret complex models, the demand for interpretability drops. The need for representation capacity (capturing patterns in high-dimensional data) grows. The Neural Network, with its ability to model non-linear relationships, becomes the compatible choice. This shift shows that the framework does not favour one model universally. It responds dynamically to the user’s evolving needs.

For the expert manager ($E = 0.99$), the Neural Network remains recommended. It has an even greater compatibility advantage. At this expertise level, interpretability is practically irrelevant. Scalability and representation capacity become primary. The Neural Network excels in both. The framework does not recommend the model with the highest theoretical capacity. It selects the one that balances the specific requirements. That underscores its practical orientation.

Practical Implications for SMEs:

These results carry implications for SMEs adopting machine learning for financial forecasting.

First, the framework provides a structured decision process. It removes guesswork from model selection. Managers do not need costly trial-and-error or intuition. They can obtain a justified, context-sensitive recommendation before training. That saves time and resources, which is critical for cash-strapped SMEs.

Second, the framework acknowledges that model selection is not one-size-fits-all. What works for a large corporation with a data science team may not suit an SME with limited expertise. By incorporating managerial expertise as a first-class variable, the framework ensures the recommended model is statistically sound and operationally feasible within the SME’s constraints.

Third, the framework encourages a nuanced view of model evaluation. It balances multiple operational requirements: interpretability, robustness, scalability, and representation capacity. These are collectively more relevant to real-world deployment. This holistic view aligns with the applied machine learning community’s consensus that success depends on human and organisational factors as much as technical performance [64, 65].

Implications for Decision Support System Design:

The validation results also offer broader insights for designing decision support systems in resource-constrained settings. The framework shows it is possible to build a lightweight recommendation engine. It does not need model fitting. It delivers contextually appropriate suggestions without heavy computational resources or historical benchmark repositories. That is valuable where cloud infrastructure is unavailable or data privacy prevents external database use.

The framework’s transparency, from explicit mathematical functions and interpretable weights, addresses a key limitation of many AutoML tools. Those often

operate as black boxes. By making the decision logic visible and explainable, the framework fosters trust. It facilitates adoption among non-technical stakeholders, who are more likely to embrace a system they can understand and scrutinise.

Summary of Validation Insights:

In summary, the Lending Club validation confirms that the framework:

1. Adapts to the manager’s expertise, recommending simpler models for non-experts and complex, high-capacity models for experts;
2. Remains consistent across compatibility metrics, indicating robust decision logic;
3. Reflects practical SME needs by balancing multiple requirements, not optimising for accuracy alone; and
4. Provides a transparent and lightweight alternative to AutoML and meta-learning, making it accessible to resource-constrained organisations.

These findings are preliminary. Further validation on additional datasets is needed. They strongly support the framework’s potential as a practical tool for context-aware model selection in SME environments. The next section discusses broader implications and outlines future research directions.

5 Discussion

5.1 Economic Implications

The economic consequences of late payments for SMEs are substantial. According to the UK Government’s Department for Business and Trade, late payments cost SMEs an average of £22,000 annually. They also contribute to about 50,000 business closures each year [64]. Research from Sage, conducted by the Centre for Economics and Business Research, reports an average overdue invoice value of about £42,000 [65]. SMEs pursue an average of 14 overdue invoices annually [66]. The UK Parliament reports an average interest rate of about 6.35% on new SME loans [67]. These figures show that payment delays are a persistent financial burden, not an isolated operational issue.

Our forecasting framework does not claim that improved predictions automatically reduce financial losses. Its main contribution is different. It supports more informed and timely managerial decisions under uncertainty. The forecasting models estimate expected payment delay. They give managers quantitative information. That information can be used in liquidity management and financing decisions. When interpreted alongside evidence on the economic costs of late payments, prediction errors (like the observed MAE values) can also be viewed for their potential managerial significance. This helps organisations prioritise follow-up actions where forecasting uncertainty may have greater financial implications. The framework’s practical value lies in reducing decision uncertainty and promoting transparent cash flow management. We do not claim direct or guaranteed financial savings.

5.2 Automated Model Selection (AutoML) and Meta-Learning

In recent years, many efforts have automated model selection and hyperparameter optimisation. We can group these efforts into four main research streams.

5.2.1 Automated Machine Learning with Hyperparameter Optimisation

Tools such as Auto-sklearn [68], Auto-WEKA 2.0 [69], and TPOT [70] combine Bayesian optimisation, genetic programming, and prior-knowledge-based hyperparameter search. They identify the best pipeline from thousands of candidates. These tools have shown strong performance in data mining competitions, such as the AutoML Challenge. They have three critical limitations for SME applications. They require multiple expensive training runs, sometimes hundreds of CPU hours. They depend heavily on cloud infrastructure and high memory capacity. They operate as black boxes. They give little explanation to non-technical managers about why a model was selected.

5.2.2 Landmarking-Based Meta-Learning

A distinct line of research, pioneered by Bensusan and Giraud-Carrier [71] and Pfahringer et al. [72], proposed a different approach. Instead of running full model training, these methods extract high-level dataset characteristics, known as meta-features. Examples include the number of instances, feature-to-instance ratio, skewness, or the performance of a few fast models (landmarkers). They compare these against a repository of past results to recommend a model. More recent work by Vanschoren [73] and Lemke and Gabrys [74] has extended this field toward similarity kernels and deep learning on meta-data.

Existing meta-learning approaches face two major challenges. Their performance depends on access to a large repository of labelled benchmark datasets. They suffer degradation when applied to new data structures, such as SME invoice records. The extraction of high-level meta-features, including landmarking, requires training at least a few baseline models. That still adds computational overhead.

5.2.3 Research Gap and Positioning of the Proposed Framework

Most existing approaches, whether heavyweight AutoML systems or repository-dependent meta-learners, frame the problem as performance optimisation or matching against historical data. None address human decision-making logic or the contextual needs of SME managers.

Our framework reframes model selection differently. It is not an expensive optimisation problem. It is a compatibility problem between operational requirements (from the business context) and intrinsic algorithmic capabilities. Three innovations stand out. The framework gives a training-free recommendation. It requires no model training. The decision process is transparent and interpretable. It explains to the manager why one model is preferred. Its computational complexity is orders of magnitude lower than conventional AutoML. You can deploy it on standard office hardware.

Given the limits of existing AutoML and meta-learning approaches in resource-constrained SME environments, we propose a lightweight and transparent framework. Section 3 provides the details.

5.3 Context-Aware Model Selection

Beyond the AutoML and meta-learning approaches discussed above, a growing body of research has explicitly addressed the problem of context-aware model selection, where the selection of a machine learning model is informed by contextual characteristics of the prediction task, such as data properties, user requirements, or environmental conditions.

5.3.1 ACCORDION

ACCORDION (Context-aware Knowledge Selection and Reliable Model Recommendation) presents a methodology that selects relevant knowledge from literature and databases to recommend models with correct structure and behaviour [75]. The framework integrates knowledge retrieval, graph algorithms, clustering, simulation, and formal analysis. ACCORDION focuses mainly on biological systems. It demonstrates the broader viability of context-aware model recommendation across domains. The authors report that their approach reduces the initial baseline model’s error by over 80%. It recommends only the most relevant subset (15% to 20%) of candidate knowledge.

5.3.2 CAMS

CAMS (Contextual Active Model Selection) tackles the problem of actively selecting pre-trained models while minimising labelling costs [76]. It is framed as an online contextual active model selection problem. CAMS uses context information to decide which model will perform best for a given context. It also has an active query component that chooses when to request labels strategically. The algorithm requires less labelling effort (under 10%) compared with existing methods on benchmark datasets. It achieves similar or better accuracy.

5.3.3 Context Aware Auto-AI Framework

The Context Aware Auto-AI Framework introduces an approach to automated machine learning [77]. It incorporates context awareness into model selection and optimisation. The framework starts with hyperparameters based on the problem type. It adapts them according to a knowledge corpus. This improves model performance and saves time and resources in tuning. The authors stress that different contexts (problem type, domain and sub-domain, input distribution changes, task requirements) can affect model performance and the tuning process significantly.

5.3.4 Context-Aware Decision Support Systems for XAI Method Selection

A decision support system has been proposed for selecting Explainable AI methods [78]. From a given set, it chooses those with higher suitability to stakeholder needs in an AI-based business problem. The approach incorporates user knowledge and feedback into an application-grounded framework. This shows how context awareness can extend beyond model selection to the selection of interpretation methods themselves.

5.3.5 Other Related Work

The literature includes context-aware recommender systems [79] and context-aware algorithm selection in specific domains. Examples include tensor factorisation [80], time-series forecasting [81], and on-device object detection [82]. This body of work shows growing recognition that model and algorithm selection must be sensitive to the operational context.

5.3.6 Positioning of the Proposed Framework

The approaches above share the broad goal of context-aware model selection. They differ from our framework in three ways.

First, with respect to **model-fitting-free recommendation**, ACCORDION, CAMS, and the Context Aware Auto-AI Framework typically require access to large knowledge bases or literature corpora. They also need online learning and labelling during deployment, or multiple training runs for optimisation. Our framework provides a training-free recommendation. It needs no model training and no external knowledge bases.

Second, regarding **transparent and interpretable decision logic**, the above frameworks often operate as black boxes. CAMS uses contextual bandits. The Auto-AI framework uses reinforcement learning. Our framework uses a transparent and coefficient-based weighting scheme derived directly from the continuous mapping functions (Section 3.3). It also uses a clear compatibility function. Every recommendation is fully traceable from contextual inputs through requirement mapping and weighting to the final assessment. Managers can see why a particular model is proposed for their context. Every step is explicit and grounded in the same mathematical functions that characterise the prediction problem. That fosters trust and helps non-technical stakeholders adopt the system.

Third, concerning **explicit integration of user expertise**, none of the above frameworks incorporate decision-maker expertise as a contextual parameter. Our framework treats user expertise as a first-class variable. It directly influences the trade-off between interpretability and accuracy. That is a critical requirement for SME applications, where non-technical managers are the primary decision-makers.

To our knowledge, no existing context-aware model selection framework combines all three features: training-free recommendation, transparent decision logic, and explicit integration of user expertise. It does so in a lightweight framework designed specifically for resource-constrained SME environments.

5.4 Limitations and Assumptions

Several limitations deserve attention. First, we used only two SME cash flow datasets: IBM Late Payment Histories and the UK Government’s Payment Practices. They cover different contexts (invoice-level and firm-level). Still, we need more validation across industries and domains to test generalisability. Second, we built the capability profiles from published characteristics. These are practical but not empirical. Future studies could estimate them from data or adapt them per domain. Third, our mapping is one possible instantiation. Other mappings, with extra variables or domain-specific requirements, might improve recommendations.

A fourth limitation concerns the profiles in Table 2. We derived them from theory and our interpretability analysis. Assigning ordinal values (0–1) to abstract constructs like interpretability and robustness involves subjectivity. Another researcher might give XGBoost a higher interpretability score because of its SHAP support. Or they might assign Neural Networks higher robustness with specific regularisation.

We mitigate this subjectivity in two ways. First, our weighting scheme does not rely on data-driven methods like CRITIC. The weights come from the mapping coefficients:

$$\alpha = 0.85, \quad \gamma = 0.70, \quad \zeta = 0.80, \quad \theta = 0.55.$$

These are fixed and theory-based. This decouples the profiles from the weighting. The recommendations stay stable across reasonable profile variations, as our sensitivity analysis shows.

Second, the framework is a customisable tool. Practitioners can adjust the profiles or weights to fit their domain or priorities. The underlying compatibility logic remains unchanged. That keeps the framework adaptable while preserving transparency and reproducibility.

We do not claim these profiles are universally optimal. They are a reproducible default. SME managers or experts can recalibrate them easily. For example, they might increase interpretability weight for regulatory needs, or lower scalability for small datasets. The theory behind the compatibility function stays intact. Future work could automate profile estimation via meta-learning, reducing subjectivity while keeping the model-fitting-free advantage.

5.5 Extensions and Future Research Directions

Several directions remain for extending the framework. Additional contextual variables could enrich the context representation. Examples include computational resource availability, regulatory constraints, data quality indicators, or organisational risk tolerance. Adaptive capability profiling and learning-based compatibility estimation could improve recommendation accuracy as more empirical evidence emerges. Evaluating the framework across a broader range of tasks would also help. These tasks include classification, anomaly detection, and time-series forecasting in different industrial domains. Such evaluations would provide evidence of robustness and applicability.

5.6 Summary and Broader Implications

In summary, this study shows that machine learning model recommendation can be a compatibility problem. It matches contextual requirements against algorithmic capabilities. This perspective moves model selection away from trial-and-error. It turns it into a transparent and structured decision process. The framework gives a practical foundation for context-aware recommendation before training. Its lightweight computation and explicit interpretability of user expertise set it apart from existing AutoML and meta-learning approaches. That makes it suitable for resource-constrained SME environments. Further validation on more datasets and domains is needed. The evidence we present supports the framework’s potential as a practical decision-support tool for context-aware model selection.

6 Conclusion

This study addressed a practical challenge: cash flow forecasting for SMEs. We used machine learning but emphasised interpretability and rigour. Much of the existing literature focuses on accuracy. It often includes future information inadvertently. We did the opposite. We imposed time-validated validation protocols. We restricted features to those available at prediction time. We evaluated models across two granularities: invoice-level transactions and firm-level aggregated attributes.

6.1 Summary of Findings

Our findings answer the three research questions directly.

First, the model with highest accuracy depends on the data context. Linear Regression and Neural Networks dominate at the invoice level. Neural Networks, Random Forest, and LightGBM cluster closely at the firm level. The accuracy gain from complex models over simpler ones is negligible in both settings. Their overfitting risk is higher, however.

Second, Random Forest gives the most actionable explanations for non-IT-expert managers. Its attributions stay consistent across methods and datasets.

Third, and most importantly, as a potential practical contribution, acknowledging that further validation is required, we propose a context-aware compatibility framework that formalises model selection as a matching problem between contextual characteristics and algorithmic capabilities. The framework translates five measurable contextual parameters, specifically data volume, noise level, data granularity, feature-to-instance ratio, and user expertise, into four operational requirements, namely interpretability, robustness, scalability, and representation capacity, through a set of continuous mathematical functions, as detailed in Section 3.3.

6.2 The Proposed Compatibility Framework as a Practical Contribution

The framework functions are calibrated on real-world SME data and grounded in established machine learning principles. They give a transparent mapping that is also theoretically defensible. This eliminates the subjectivity found in ad hoc heuristics.

The relative importance of each operational requirement comes from a coefficient-based weighting scheme. That scheme is derived directly from the same functions. We do not rely on data-dependent heuristics like CRITIC or idea-based systems like AHP. This ensures the recommendations stay stable and context-driven. Each candidate model has a predefined capability profile across four dimensions. A compatibility function picks the model whose profile best matches the contextual requirements. Sensitivity analysis supports the framework’s robustness. It confirms that the recommendation logic stays consistent under reasonable variations in the mapping parameters.

We validated the accuracy and interpretability on two independent SME cash flow datasets: the IBM Late Payment Histories (invoice level) and the UK Government’s Payment Practices dataset (firm level). We also did a validation on the Lending Club dataset to check our framework in action. The experimental results show that the framework gives context-sensitive recommendations. They shift logically with managerial expertise. Non-expert users get interpretable models. Technical experts get high-capacity models. This behaviour matches the performance of the candidate models. It confirms that the framework operationalises the trade-off between interpretability and predictive power effectively.

We must stress that this framework is an exploratory proof of concept. Independent validation on larger and more diverse datasets, including the SDF dataset [83], can be discussed in our future work. Future work will also explore automated estimation of capability profiles. We will extend the framework to other tasks, such as classification and anomaly detection.

6.3 Theoretical and Practical Implications

From a theoretical angle, our results add to the evidence that algorithm selection in financial machine learning should be sensitive to data granularity and user constraints. We should not assume a single best model for all settings. From a practical standpoint, we translate these findings into actionable guidelines. They are domain-aligned. Accelerate e-invoicing adoption at the invoice level. Renegotiate contractual payment periods at the firm level. Both actions improve liquidity.

6.4 Final Remarks

For many cash-constrained SMEs, a model that is transparent and interpretable may offer more practical value than an opaque alternative with only a tiny accuracy gain. This study shifts the process from trial-and-error to a context-aware compatibility check. It gives a practical and clear basis for recommending machine learning models before you start building them.

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Declarations

Competing Interests

The authors declare that they have no competing interests.

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Data Availability

The datasets analysed during the current study are publicly available:

- IBM Late Payment Histories: <https://www.kaggle.com/datasets/hhenry/finance-factoring-ibm-late-payment-histories>
- UK Government's Payment Practices: <https://www.kaggle.com/datasets/saikiran0684/payment-practices-of-uk-buyers>
- Lending Club Loan Data: <https://www.kaggle.com/datasets/wordsforthewise/lending-club>

Code Availability

The Python implementation of the proposed framework is available at <https://github.com/aliNzLami/cashFlow-forecasting-ML>.

Ethics Approval and Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Supplementary Information

The Python implementation of the proposed context-aware compatibility framework, along with all experimental scripts, data preprocessing pipelines, and analysis modules, is publicly available at the following GitHub repository:

Repository URL: <https://github.com/aliNzLami/cashFlow-forecasting-ML>

The repository contains the following materials:

- Complete implementation of the context-to-requirement mapping functions, compatibility scoring, and recommendation algorithm;

- Scripts for reproducing all experimental results reported in this paper, including accuracy and interpretability analyses;
- Preprocessing pipelines for the IBM Late Payment Histories, UK Government’s Payment Practices, and Lending Club datasets;
- Calibration and sensitivity analysis modules;
- Detailed README documentation with installation instructions and usage examples.

References

- [1] Zemmouchi-Ghomari, L. and Maroua, M. (2025). Beyond failure rates: unveiling startup success factors with machine learning. *Journal of Computational Social Science*, 8(3), Article 15. doi:10.1007/s42001-025-00398-4. Available at: <https://link.springer.com/article/10.1007/s42001-025-00398-4>
- [2] Blakeman, C. (2025). The One Reason Most Small Businesses Fail. And No, It Isn’t Money. *Entrepreneur*, 14 March. Available at: <https://www.entrepreneur.com/leadership/the-one-reason-most-small-businesses-fail-and-no-it-isnt/487771>
- [3] CB Insights. (2021). The Top 12 Reasons Startups Fail [Report]. Available at: https://www.cbinsights.com/reports/CB-Insights_Top-Reasons-Startups-Fail.pdf
- [4] Malkus, B., Bobek, S. and Nalepa, G.J. (2025). Financial Management System for SMEs: Real-World Deployment of Accounts Receivable and Cash Flow Prediction. *arXiv*, arXiv:2511.03631. Available at: <https://arxiv.org/abs/2511.03631>
- [5] IEEE. (2025). Literature Review on Optimizing Cash Flow Forecasting Using Machine Learning in Small and Medium Enterprises. In: *2025 International Conference on...*, Prawet, Thailand, 10-12 March. IEEE. doi:10.1109/10968322. Available at: <https://ieeexplore.ieee.org/document/10968322>
- [6] Ida, M., Kai, I. and Kondo, S. (2026). — [Construction of a prediction model for significant deterioration of operating cash flow: Potential for reducing financial institutions’ credit-related costs]. *JCER Data Science Research Paper*. Japan Center for Economic Research. Available at: <https://www.jcer.or.jp/research-report/20260318-4.html>
- [7] Kanatuly, S. N. and Zhankozhauly, K. A. (2026). Cash Flow Forecasting Using Time Series Analysis, Machine Learning and Deep Learning Models: Evidence from New York City Financial Data. *Universum*, 4(145). doi: 10.32743/UniTech.2026.145.4.22565. Available at: <https://cyberleninka.ru/article/n/cash-flow-forecasting-using-time-series-analysis-machine-learning-and-deep-learning-models-evidence-from-n>

- [8] de Lima, S. M. L. (2025). Time Series Forecasting with Exogenous Variables: A Literature Review to Identify Promising Gaps in Computational Research. *Revista Principia*, 62. doi: 10.18265/2447-9187a2025id8575. Available at: <https://doaj.org/article/80242ed475594592878be36849a89bb7>
- [9] Phuong, N. T. T., Hang, N. T., Thanh, N. T., and Ngoc, B. T. (2024). Impact of Financial Ratios on Net Cash Flow of Vietnamese Listed Companies. In: *Proceedings of the 4th International Conference on Research in Management and Technovation*, T. H. N. Nguyen, D. N. Burrell, V. K. Solanki, and N. A. Mai, Eds. Singapore: Springer. doi: 10.1007/978-981-99-8472-5_52. Available at: https://link-proxy.springer.com/chapter/10.1007/978-981-99-8472-5_52
- [10] Do Accruals Convey Information About Future Cash Flows? A Re-Examination of Inferences Drawn. (2026). *Journal of Business Finance & Accounting*. doi: 10.1111/jbfa.70068. Available at: <https://onlinelibrary.wiley.com/doi/full/10.1111/jbfa.70068>
- [11] A nonlinear dynamic approach to cash flow forecasting. (2022). *Springer-Professional*, 16 May. Available at: <https://www.springerprofessional.de/a-nonlinear-dynamic-approach-to-cash-flow-forecasting/20419418>
- [12] Pfahringer, B., Bensusan, H., & Giraud-Carrier, C. G. (2000). Meta-learning by landmarking various learning algorithms. *Proceedings of the International Conference on Machine Learning (ICML)*, 743-750. Available at: <https://dblp.org/rec/conf/icml/PfahringerBG00>
- [13] Vanschoren, J. (2018). Meta-learning: A survey. *arXiv*, arXiv:1810.03548. Available at: <https://arxiv.org/abs/1810.03548>
- [14] Lemke, C., & Gabrys, B. (2010). Meta-learning for time series forecasting and forecast combination. *Neurocomputing*, 73(10-12), 2006-2016. doi: 10.1016/j.neucom.2009.09.020. Available at: <https://doi.org/10.1016/j.neucom.2009.09.020>
- [15] Xing, Q., Yu, C., Huang, S., Zheng, Q., Mu, X., and Sun, M. (2024). Credit Score Prediction Using Ensemble Model. *arXiv*, arXiv:2410.00256. Available at: <https://arxiv.org/html/2410.00256>
- [16] Ahmed, Y., Telmer, C. A., Zhou, G., & Miskov-Zivanov, N. (2024). Context-aware knowledge selection and reliable model recommendation with ACCORDION. *Frontiers in Systems Biology*. Available at: <https://www.frontiersin.org/journals/systems-biology/articles/10.3389/fsysb.2024.1308292/full>
- [17] Mehra, S., Aggarwal, S., and Mitra, P. (2024). Predictive Analytics and AI-Driven Strategies for Enhanced Cash Flow Forecasting. In: *Lecture Notes in Networks and Systems*, vol. 1065 LNNS, Springer, pp. 296-315. doi: 10.1007/978-3-031-66329-1_21. Available at: <https://www.mendeley.com/>

catalogue/bf3356f2-90a1-33a4-af0b-5d115eb0fcd6/

- [18] Vu, M.-C., Dinh, T. L., Le, T. D., and Nguyen, T. L. H. (2025). A Conceptual Model for AI Adoption in Financial Decision-Making: Addressing the Unique Challenges of Small and Medium-Sized Enterprises. *arXiv*, arXiv:2512.04339. Available at: <https://arxiv.org/abs/2512.04339>
- [19] Context based time series analysis and forecasting: A review of methods and applications. (2025). *AIP Publishing*. Available at: <https://pubs.aip.org>
- [20] Fernaldy, E., Santy, D., & Deniawara, K. (2025). Literature review on optimizing cash flow forecasting using machine learning in SMEs. *ICMLAS, IEEE*. Available at: <https://ieeexplore.ieee.org/document/10968322>
- [21] Malakauskas, A., & Lakstutiene, A. (2021). Financial distress prediction for SMEs using machine learning. *Inzinerine Ekonomika-Engineering Economics*, 32(1). doi: 10.5755/j01.ee.32.1.27382. Available at: <https://doi.org/10.5755/j01.ee.32.1.27382>
- [22] Małkus, B., Bobek, S., & Nalepa, G. J. (2026). Financial management system for SMEs: Real-world deployment. *arXiv*, arXiv:2511.03631. Available at: <https://arxiv.org/abs/2511.03631>
- [23] Sharmetov, N., & Kartbayev, A. (2026). Cash flow forecasting using time series, machine learning and deep learning. *Universum*. Available at: <https://cyberleninka.ru/article/n/cash-flow-forecasting-using-time-series-analysis-machine-learning-and-deep-learning-models-evidence-from-n>
- [24] Kureljusic, M., & Metz, J. (2023). Applicability of ML algorithms in accounts receivables management. *Journal of Applied Accounting Research*. doi: 10.1108/JAAR-05-2022-0116. Available at: <https://doi.org/10.1108/JAAR-05-2022-0116>
- [25] Kazar, G. (2024). Data-driven approach for predicting cash flow performance. *Engineering, Construction and Architectural Management*. doi: 10.1108/ECAM-06-2024-0706. Available at: <https://doi.org/10.1108/ECAM-06-2024-0706>
- [26] Meta-Learning Enhanced AutoML for Multi-Objective Optimization in Small and Medium-Sized Enterprises. (2025). *IEEE Xplore*. Available at: <https://ieeexplore.ieee.org/document/11188213>
- [27] Meta-Learning Enhanced AutoML for Multi-Objective Optimization in Small and Medium-Sized Enterprises. (2025). *IEEE Xplore*. Available at: <https://ieeexplore.ieee.org/document/11188213>
- [28] Mangold, A., Zietz, J., Weinhold, S., and Pannasch, S. (2025). On the Design and Evaluation of Human-Centered Explainable AI Systems: A Systematic Review

- and Taxonomy. *arXiv*, arXiv:2510.12201. Available at: <https://arxiv.org/pdf/2510.12201v1#8#1>
- [29] Zein, A. (2025). Personalized Explainable AI: Dynamic Adjustment of Explanations for Novice and Expert Users. *J. PIPA*, 11(9). doi: 10.29303/jppipa.v11i9.12586. Available at: <https://jppipa.unram.ac.id/index.php/jppipa/article/view/12586>
 - [30] Noise Tolerance and Robustness Ranking in Machine Learning Models. (2024). In: *Proc. IDEAL 2024*, LNCS, vol. 15347, pp. 96–108. Available at: https://dtic.dimensions.ai/details/publication/pub.1182525506?and_facet_funder=grid.482851.2&and_facet_for=80017&and_facet_sdg=40013
 - [31] Prokopchina, S. V. and Zvyagin, L. S. (2025). Intelligent information processing technologies for managing small and medium-sized enterprises based on a regularizing bayesian approach. *Comp. nanotechnol.*, 12(4), pp. 40–50. doi: 10.33693/2313-223X-2025-12-4-40-50. Available at: https://m.mathnet.ru/php/archive.phtml?wshow=paper&jrnid=cn&paperid=591&option_lang=eng
 - [32] Large-Scale Bankruptcy Risk Prediction for Romanian SMEs Using Robust Machine Learning Approaches. (2025). In: *Proc. 27th Int. Symp. Symbolic Numer. Algorithms Sci. Comput. (SYNASC)*. Available at: <https://www.computer.org/csdl/proceedings-article/synasc/2025/901100a355/2fRJekakpcQ>
 - [33] Islam, M., Sharma, A., and Khilji, N. (2025). Utilising Machine Learning Algorithms to Address Computational Challenges in Big Data Analytics. *Int. J. Comput. Eng. Sci. Eng. Netw.* doi: 10.22399/ijcesen.3165. Available at: <https://www.ijcesen.com/index.php/ijcesen/article/view/3165>
 - [34] Scalable Artificial Intelligence Models for Big Data Processing and Analysis. (2026). In: *Proc. Int. Conf. Artif. Intell. Comput. Eng. (ICAICE)*, Erode, India, Mar. 2026. Available at: <https://ieeexplore.ieee.org/document/11502904>
 - [35] Effects of multicollinearity and data granularity on regression models of stream temperature. (2024). *J. Hydrol.*, 638, 131498. doi: 10.1016/j.jhydrol.2024.131498. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0022169424009685>
 - [36] Sabharwal, R., Miah, S. J., Wamba, S. F., and Cook, P. (2024). Extending application of explainable artificial intelligence for managers in financial organizations. *Ann. Oper. Res.* doi: 10.1007/s10479-024-05825-9. Available at: <https://www.mendeley.com/catalogue/86ff97a5-ea26-3a04-a42d-32acc8973061/>
 - [37] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 30. Available at: <https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf>

- [38] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD*, 1135-1144. doi: 10.1145/2939672.2939778. Available at: <https://doi.org/10.1145/2939672.2939778>
- [39] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5-32. doi: 10.1023/A:1010933404324. Available at: <https://doi.org/10.1023/A:1010933404324>
- [40] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD*, 785-794. doi: 10.1145/2939672.2939785. Available at: <https://doi.org/10.1145/2939672.2939785>
- [41] Ke, G., Meng, Q., Finley, T., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems (NeurIPS)*, 30. Available at: <https://papers.nips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree>
- [42] Zabërgja, G., Kadra, A., Frey, C. M. M., & Grabocka, J. (2024). Tabular Data: Is Deep Learning all you need? *arXiv*, arXiv:2402.03970. Available at: <https://arxiv.org/abs/2402.03970>
- [43] Henry, H. (2024). Finance Factoring - IBM Late Payment Histories [Dataset]. *Kaggle*. Available at: <https://www.kaggle.com/datasets/hhenry/finance-factoring-ibm-late-payment-histories>
- [44] Malkus, B. and Nalepa, G.J. (2022). Forecasting Daily Cash Flows in a Company - Shortcoming in the Research Field and Solution Exploration. In: *Machine Learning, Optimization, and Data Science*. Springer, Cham, pp. 377-391. doi: 10.1007/978-3-031-25599-1_27. Available at: https://dl.acm.org/doi/10.1007/978-3-031-25599-1_27
- [45] Saikiran V. (2017). Payment Practices of UK Buyers [Dataset]. *Kaggle*. Available at: <https://www.kaggle.com/datasets/saikiran0684/payment-practices-of-uk-buyers>
- [46] UK Government. (2017). The Reporting on Payment Practices and Performance Regulations 2017. *Statutory Instrument 2017 No. 395*. Available at: <https://www.legislation.gov.uk/uksi/2017/395/contents/made>
- [47] UK Government. (2026). Large businesses' payment practices and performance: Statistics (2025). Available at: <https://www.gov.uk/government/statistics/announcements/large-businesses-payment-practices-and-performance-statistics-2025>
- [48] Lending Club. (2018). Lending Club Loan Data [Dataset]. *Kaggle*. Available at: <https://www.kaggle.com/datasets/wordsforthewise/lending-club>

- [49] Baumöhl, E., Lyócsa, Š., and Vašaničová, P. (2024). Macroeconomic environment and the future performance of loans: Evidence from three peer-to-peer platforms. *Int. Rev. Financ. Anal.*, 95, 103416. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S105752192400348X>
- [50] Cheng, T., Jiang, S., Zhao, A.B. and Zhao, J. (2025). Is machine learning a necessity? A regression-based approach for stock return prediction. *Journal of Empirical Finance*, 81, 101598. doi: 10.1016/j.jempfin.2025.101598. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0927539825000209>
- [51] Wang, Y. and Chen, X. (2026). Soft probability based random forest for financial distress prediction. *Information Sciences*, 729, 122870. doi: 10.1016/j.ins.2025.122870. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0020025525010060>
- [52] Li, J. and Zhang, H. (2026). Frequency-aware gradient modulated boosted trees for interpretable financial distress prediction. *Chaos, Solitons & Fractals*. doi: 10.1016/j.chaos.2026.116342. Available at: <https://ideas.repec.org/a/eee/chsofr/v208y2026ip2s0960077926002948.html>
- [53] Zhou, M. and Liu, Y. (2026). Interpretable AI for financial risk perception: A machine learning approach to corporate crisis prediction. *Finance Research Letters*, 89, 109317. doi: 10.1016/j.frl.2025.109317. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S1544612325025668>
- [54] Berger, T. (2025). Deep Learning and Econometric Time Series Analysis: An Assessment of Daily Return Forecasts. *Journal of Forecasting*. doi: 10.1002/for.70045. Available at: <https://onlinelibrary.wiley.com/doi/full/10.1002/for.70045>
- [55] Beyond One-Size-Fits-All: Comparing and Selecting Regression Metrics for Robust Model Assessment. (2025). In: *2025 12th International Conference on Information Technology (ICIT)*, Amman, Jordan, 2025, pp. 416–422. doi: 10.1109/ICIT64950.2025.11049268. Available at: <https://ieeexplore.ieee.org/document/11049268>
- [56] Chai, T. and Draxler, R.R. (2014). Root mean square error (RMSE) or mean absolute error (MAE)? – Arguments against avoiding RMSE in the literature. *Geoscientific Model Development*, 7(3), pp. 1247-1250. doi: 10.5194/gmd-7-1247-2014. Available at: <https://www.geosci-model-dev.net/7/1247/2014/>
- [57] Hyndman, R.J. and Koehler, A.B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), pp. 679-688. doi: 10.1016/j.ijforecast.2006.03.001. Available at: <https://ehl.robjhyndman.com/publications/another-look-at-measures-of-forecast-accuracy/>

- [58] Lundberg, S.M. and Lee, S.I. (2017). A Unified Approach to Interpreting Model Predictions. In: *Advances in Neural Information Processing Systems (NeurIPS)*, 30, pp. 4765-4774. Available at: <https://arxiv.org/abs/1705.07874>
- [59] Saabas, A. (2016). Interpreting Random Forests with treeinterpreter. Available at: <https://blog.datahive.net/interpreting-random-forests/>
- [60] Ribeiro, M.T., Singh, S. and Guestrin, C. (2016). Why Should I Trust You? Explaining the Predictions of Any Classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135-1144. Available at: <https://dl.acm.org/doi/10.1145/2939672.2939778>
- [61] Shrikumar, A., Greenside, P. and Kundaje, A. (2017). Learning Important Features Through Propagating Activation Differences. *arXiv*, arXiv:1704.02685. Available at: <https://arxiv.org/abs/1704.02685>
- [62] Kokhlikyan, N. et al. (2020). Captum: A unified and generic model interpretability library for PyTorch. *arXiv*, arXiv:2009.07896. Available at: <https://arxiv.org/abs/2009.07896>
- [63] A. N. Lamiry. Cash Flow Forecasting Repository. *GitHub*, Available at: <https://github.com/aliNzLami/cashFlow-forecasting-ML>
- [64] UK Government. (2024). Crack down on late payments in major support package for small businesses. Available at: <https://www.gov.uk/government/news/crack-down-on-late-payments-in-major-support-package-for-small-businesses>
- [65] Sage. (2025). Addressing late payments could unlock £112 Billion in cashflow for small businesses. Available at: <https://www.sage.com/en-gb/company/digital-newsroom/2025/05/14/addressing-late-payments-could-unlock-112-billion-in-cashflow-for-small-businesses/>
- [66] SME Today. (2026). Small businesses owed over £12k in late payments every year, report finds. Available at: <https://www.smetoday.co.uk/finance/small-businesses-owed-over-12k-in-late-payments-every-year-report-finds/>
- [67] UK Parliament. (2025). Written question 92507: Interest rates for small and medium sized enterprises. Available at: <https://questions-statements.parliament.uk/written-questions/detail/2025-11-19/92507>
- [68] Feurer, M., Klein, A., Eggenberger, K., Springenberg, J., Blum, M., & Hutter, F. (2015). Efficient and robust automated machine learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 28. Available at: <https://papers.nips.cc/paper/5872-efficient-and-robust-automated-machine-learning.pdf>
- [69] Kotthoff, L., Thornton, C., Hoos, H. H., Hutter, F., & Leyton-Brown, K. (2016). Auto-WEKA 2.0: Automatic model selection and hyperparameter optimization

- in WEKA. *Journal of Machine Learning Research*, 17(1), 825-830. Available at: <https://jmlr.csail.mit.edu/papers/v17/15-599.html>
- [70] Olson, R. S., & Moore, J. H. (2016). TPOT: A tree-based pipeline optimization tool for automating machine learning. *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO)*, 66-74. doi: 10.1145/2908812.2908918. Available at: <https://doi.org/10.1145/2908812.2908918>
 - [71] Bensusan, H., & Giraud-Carrier, C. (2000). Discovering task neighbourhoods through landmark learning performances. *European Conference on Principles of Data Mining and Knowledge Discovery (PKDD)*, 325-330. Available at: https://link.springer.com/chapter/10.1007/3-540-45372-5_32
 - [72] Pfahringer, B., Bensusan, H., & Giraud-Carrier, C. G. (2000). Meta-learning by landmarking various learning algorithms. *Proceedings of the International Conference on Machine Learning (ICML)*, 743-750. Available at: <https://dblp.org/rec/conf/icml/PfahringerBG00>
 - [73] Vanschoren, J. (2018). Meta-learning: A survey. *arXiv*, arXiv:1810.03548. Available at: <https://arxiv.org/abs/1810.03548>
 - [74] Lemke, C., & Gabrys, B. (2010). Meta-learning for time series forecasting and forecast combination. *Neurocomputing*, 73(10-12), 2006-2016. doi: 10.1016/j.neucom.2009.09.020. Available at: <https://doi.org/10.1016/j.neucom.2009.09.020>
 - [75] Ahmed, Y., Telmer, C. A., Zhou, G., & Miskov-Zivanov, N. (2024). Context-aware knowledge selection and reliable model recommendation with ACCORDION. *Frontiers in Systems Biology*. Available at: <https://www.frontiersin.org/journals/systems-biology/articles/10.3389/fsysb.2024.1308292/full>
 - [76] Liu, X., Xia, F., Stevens, R. L., & Chen, Y. (2022). Contextual Active Model Selection. *arXiv*, arXiv:2207.06030. Available at: <https://arxiv.org/abs/2207.06030>
 - [77] Context Aware Auto-AI Framework. (2024). In: *Springer, International Conference on Data Science and Artificial Intelligence*. Available at: https://rd.springer.com/chapter/10.1007/978-3-031-53963-3_44
 - [78] Reis, M. I., Gonçalves, J. N. C., Cortez, P., Carvalho, M. S., and Fernandes, J. M. (2025). A context-aware decision support system for selecting explainable artificial intelligence methods in business organizations. *Comput. Ind.*, 165, 104233. doi: 10.1016/j.compind.2024.104233. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0166361524001611?via%3Dihub>
 - [79] Le, Q., Vu, S., Nguyen, T., and Le, T. (2021). A State-of-the-Art Survey on Context-Aware Recommender Systems and Applications. *Int. J. Knowl. Syst.*

- Sci.*, 12(3), pp. 1–20. doi: 10.4018/IJKSS.2021070101. Available at: <https://www.igi-global.com/gateway/article/285959>
- [80] Cunha, T., Soares, C., and de Carvalho, A. C. P. L. F. (2017). Metalearning for Context-aware Filtering: Selection of Tensor Factorization Algorithms. In: *Proceedings of the Eleventh ACM Conference on Recommender Systems (RecSys '17)*, Aug. 2017, pp. 384–385. doi: 10.1145/3109859.3109899. Available at: <https://dl.acm.org/doi/10.1145/3109859.3109899>
 - [81] Sousa, J. and Henriques, R. (2026). T2f: Actor-critic reinforcement learning for time-series forecasting. *Neural Comput. Appl.*, 38(12), 510. doi: 10.1007/s00521-026-12209-6. Available at: <https://link.springer.com/article/10.1007/s00521-026-12209-6>
 - [82] Kang, S., Jeong, C., and Chung, K. (2021). Context-aware Model Selection for On-Device Object Detection. In: *2021 International Conference on Information Networking (ICOIN)*, Jan. 2021, pp. 662–666. doi: 10.1109/ICOIN50884.2021.9333918. Available at: <https://ieeexplore.ieee.org/document/9333918/>
 - [83] Baillie, S. (2026). New research-ready datasets available in myFoundry. *Smart Data Foundry*, Feb. 12, 2026. Available at: <https://smartdatafoundry.com/resources/blog/new-research-ready-datasets-available-in-myfoundry>